How Language Models Access Information Beyond the Local Context:

A Tutorial on Long Context, Retrieval, Structured Indexing, Search Agents, and Progressive Retrieval Attention

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August 2026

Abstract

Language models can access information beyond their local token stream in several ways: they can enlarge or sparsify attention, carry recurrent state, update neural memory at test time, retrieve text or latent representations, compress and reuse K/V caches, navigate an index, or call tools. These mechanisms are usually studied in separate literatures. We ask a common question instead: *what access contract connects the model to information that is not already local?* An access contract specifies what is stored, how an item is identified, when selection occurs, which representation is materialized, how it enters the model, and which intervention would demonstrate that the selected information caused an output change.

This tutorial develops that comparison across seven architectural families. Progressive Retrieval Attention (PRAttention, PRA) serves as a worked case rather than a presumed winner. Its corrected prototype represents external documents as explicit URI-addressed references, partitions each reference into chunks, derives one contextual routing gist per chunk and layer, and transports full token K/V only for selected chunks. We reconstruct this design, distinguish it from prior retrieval and memory mechanisms, and mark its earlier reference-conditioned experiments as historical because they used a superseded router. The survey then derives matched comparisons and counterfactual controls for testing long context, retrieval, latent memory, structural navigation, and agents on common quality and cost frontiers.

Contents

1 Introduction

The canonical Transformer converts a token sequence into contextual representations using dense self-attention. Its generality comes with an implicit boundary: the model can use only information that has already been placed in the sequence or encoded in its parameters. Long-document and knowledge-intensive tasks violate this boundary in different ways. A source may be too large to fit, only a few passages may matter, evidence may be nested inside a document hierarchy, or the identity of the useful source may emerge only after partial reasoning.

Consider a model repairing a software repository. The repository contains thousands of files, but the immediate question concerns one failing authentication test. A dense-context system loads many files eagerly. A sparse transformer loads the same tokens but changes their interaction graph. A retriever inserts a few passages before generation. A structural navigator follows repository, module, class, and method nodes. An agent searches, opens files, runs tests, and revises its plan. A recurrent or latent-memory model may instead carry information from earlier inspection. These systems do not merely use different algorithms; they place the boundary between local computation and external information in different locations.

This observation motivates the survey's organizing question. Sparse attention changes the graph after source text is local. Recurrence preserves state across segments. Differentiable memory stores learned or episodic representations. Retrieval-augmented generation selects passages and usually materializes them as text. Attention-level retrieval inserts hidden states or K/V closer to the decoder. Structural indexes preserve document organization. Search agents acquire information through explicit actions. PRA proposes another contract: a compact local handle names an object, while selection, resolution, materialization, and fusion remain separate operations. The purpose of the taxonomy is to compare these contracts, not to declare one universally preferable.

The review proceeds from a shared vocabulary to one worked architecture and then to the broader field. Section 2 reconstructs PRA in detail because its design exposes all parts of the access contract in one system. Sections 3–9 then examine the seven families at their own system boundaries. The landscape and comparison table synthesize those accounts, after which the experimental sections turn architectural differences into matched tests. The final sections develop research extensions and limitations. This order is pedagogical rather than evaluative: presenting PRA first does not assign it a higher evidential status than the established methods that follow.

1.1 Seven Access Families

The seven categories below follow the location at which an architecture changes the information path. They are not mutually exclusive. A practical system may use sparse local attention, a retrieval index, a K/V cache, and an agent escalation policy at the same time.

1. **Dense, sparse, and approximate long-context attention**: extend the visible token graph.
2. **Recurrence and compression**: carry activations, latent slots, associative state, or test-time neural memory across segments.
3. **Differentiable external memory**: read and write learned memory slots or episodic stores.
4. **Retrieval-augmented pretraining and generation**: retrieve textual evidence before or during generation.
5. **Attention-level and KV retrieval**: select, retain, compress, allocate, or reuse hidden-state and key/value memories near attention.
6. **Structural and hierarchical indexing**: navigate document, graph, or summary structure rather than flat chunks.
7. **Agents with search tools**: iteratively formulate queries, inspect results, follow links, and decide when to stop.

1.2 A Shared Vocabulary

The repository example suggests a decomposition that applies across categories. First, a name or index entry says what information is available; this may be a URI, document identifier, token position, or search result. The model then has a local state, such as the hidden representation of the failing test. A policy chooses candidates, perhaps an authentication module or a prior incident. The system materializes those candidates as source tokens, summaries, hidden states, K/V tensors, graph nodes, or tool observations. Finally, a fusion mechanism makes the representation available to the model. A compact item used to choose evidence should not be confused with the detailed evidence supplied after that choice.

We can express this process without committing to a particular architecture. Let $x_{1:T}$ denote the local tokens and let h_t be the model state at generation step t. A query q_t is derived from that state. The external store is $\mathcal{S}$, and R

is a selection policy with configuration π. The selected candidates are

$$C_t = R(q_t, \mathcal{S}; \pi), \tag{1}$$
$$M_t = \text{materialize}(C_t; \rho), \tag{2}$$
$$h'_t = \text{combine}(h_t, M_t; \gamma), \tag{3}$$

where C_t is the candidate set. The materialization policy ρ determines whether those candidates become tokens, latent states, K/V tensors, or observations. The fusion policy γ determines whether M_t is concatenated to a prompt, read by cross-attention, injected into a cache, or processed by another model call. This vocabulary separates three questions that are often conflated: whether the right information was selected, whether its representation preserved the needed evidence, and whether the model used that representation. The next section applies the decomposition to PRA before the survey compares it with the other six families.

1.3 How to Read the Comparisons

Each method account follows the same sequence. It begins with the pressure that motivated the method, gives an intuition and formal mechanism, traces the resulting data flow, and then discusses evidence and failure modes. The final paragraph asks how to compare that method with PRA without changing several variables at once. These proposed comparisons are experimental designs, not reports of completed experiments.

Claims also have different sources. Statements about prior systems are supported by the cited primary work. Statements about the corrected PRA prototype describe repository behavior that has been implemented and tested mechanically. Quantitative PRA results come from the superseded v1 router unless identified as ordinary-language backbone results. Predictions about reuse, recursive benefit, learned selection, or favorable quality–cost trade-offs remain hypotheses. Keeping these statuses visible prevents a software capability from being read as empirical validation.

2 Progressive Retrieval Attention

2.1 PRA as a Worked Access Contract

PRA is easiest to understand as an *access contract*, not as a claim that another attention formula solves long context. Consider the repository example from the introduction. A token such as `<REF_1>` can name the authentication module without placing that module's full source in the prompt. The name identifies a possible source; it does not decide whether to read the source or how to represent it. Paper 0 generalizes this observation into a logical context graph. Paper 1 specializes it into a small decoder-only PyTorch model. Together they separate five operations:

1. **naming**: a handle states which object may be consulted;
2. **selection**: a policy chooses among advertised objects;
3. **resolution**: a resolver binds a URI, policy, and version to content;
4. **materialization**: content becomes text, a summary, hidden states, or layer-specific keys and values; and
5. **fusion**: the decoder consumes that representation through prompt tokens, cross-attention, native KV injection, or another transport.

The list summarizes a causal chain rather than five interchangeable names. A better selector does not establish a better transport, and a useful memory branch does not show that the selector chose the correct object. PRA's proposed package combines explicit URI identity, separate reference and chunk routing, reusable per-layer K/V objects, progressive triggering, and counterfactual evaluation. Each component has precursors. The scientific question is whether this combination improves quality per unit of active context, latency, and memory in regimes with sparse, reusable, naturally named evidence.

At the representation level, the same chain has three easy-to-confuse objects. A handle is a symbolic address. A gist is a small, layer-specific index entry used to rank one chunk. Detailed token K/V is the material the decoder reads after selection. The prototype can compute the latter two ahead of the prompt forward or recover them from a cache; this scheduling choice does not erase their different roles.

2.2 Reference graph, resolver, and runtime state

The repository example becomes a graph when a module refers to a class, the class refers to a method, and the method refers to a configuration object. Let this context graph be $\mathcal{G} = (\mathcal{V}, \mathcal{E})$. A node $v = (u, z, s, m, R)$ contains a stable URI u, content z, optional summary metadata s, metadata m, and a local reference table R. An edge (v_i, a, v_j) exists when $R_i[a] = u_j$; lexical URI prefixes do not create edges implicitly. The resolver remains outside

the neural layer:

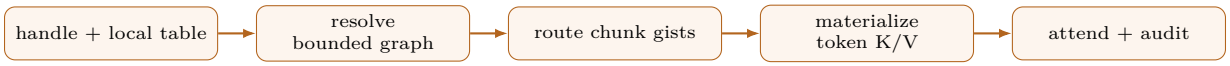
$$\rho(u, p, t) \longrightarrow (z, s, R, m, \nu), \tag{4}$$

where p is an authorization domain, t identifies logical time, and ν is the resolved version. This interface permits a file, database row, document section, code symbol, graph query, or tool result to share a naming scheme without pretending that they share a physical representation.

The runtime state before token t is

$$\Omega_t = (\mathcal{G}, \mathcal{H}_t, A_t, \mathcal{C}_t, p_t, \nu_t), \tag{5}$$

with advertised handles $\mathcal{H}_t$, active references A_t, resident materializations $\mathcal{C}_t$, policy state p_t, and provenance/version state ν_t. A persistent cache key must include at least $(u, v, \ell, m, \theta, p)$: URI, content version, consumer layer, model identity, relevant encoder parameters, and policy domain. Paper 1 uses only the URI because it rebuilds and discards a private cache for every example; that specialization prevents cross-example leakage but does not demonstrate persistent reuse.



2.3 Selection, recursive expansion, and the implemented specialization

Selection should be cheap enough to avoid materializing every candidate, but it must use a representation compatible with the consuming layer. The corrected prototype therefore partitions reference u into chunks $c \in \mathcal{C}_u$. At layer ℓ, it reconstructs the projected key heads for every chunk token and mean-pools them into one routing gist:

$$g^K_{u,c,\ell} = \frac{1}{|c|} \sum_{j \in c} \text{mergeheads}(K_{u,c,\ell,j}). \tag{6}$$

The current layer's projected last-token query is

$$q_{t,\ell} = \text{mergeheads}(X^{(\ell)} W^{(\ell)}_Q)_t, \tag{7}$$

and its content-routing score for chunk c is

$$a_{u,c,\ell} = \cos(q_{t,\ell}, g^K_{u,c,\ell}). \tag{8}$$

The default hierarchical policy aggregates chunk scores into a reference score, selects at most k_r references, and independently selects at most k_c chunks within each retained reference. Optional summaries can replace, compete with, or augment the content score, but they are disabled by default and missing summaries fall back to content. Thus the last local query first chooses a small working set. It does not yet read the selected passages, and its cosine scores are not the later token-level attention map.

An expansion operator $\mathcal{X}(A, \mathcal{G}, b, d)$ follows explicit child tables under shared branching, reference, token, and depth budgets. The resulting transition is

$$A_t = \mathcal{S}_{k,\tau}(q_t, \mathcal{C}_t), \qquad A'_t = \mathcal{X}(A_t, \mathcal{G}, b, d), \tag{9}$$

$$\mathcal{C}'_t = \text{admit}(\text{encode}(\rho(A'_t))), \qquad (y_t, \Omega_{t+1}) = \text{attend}(x_{\le t}, \mathcal{C}'_t, \Omega_t). \tag{10}$$

The runtime now implements deterministic child-first construction, explicit build states, cycle policies, and shared budgets. Parent encoding may read already completed child memory. Concretely, if a module points to a policy object, the policy object is made ready before the parent module is encoded; the parent can then form its own layer states with that child memory available. Cycle markers and shared budgets keep this procedure from turning a graph into unbounded recursion. This establishes bounded recursive mechanics, not a learned expansion policy or an empirical recursion benefit. Persistent cross-request reuse also remains unmeasured.

2.4 Layer-specific materialization and PRAttention

Routing gists identify promising chunks; they are not the evidence supplied to the decoder. For URI u, the cache entry instead stores full token memory and one gist for every chunk and PRA layer:

$$C_u = \left(u, z_u, m_u, \{(c, K_{u,c,\ell}, V_{u,c,\ell}, g^K_{u,c,\ell}, g^V_{u,c,\ell})\}_{c,\ell}\right). \tag{11}$$

Here $K_{u,c,\ell}$ and $V_{u,c,\ell}$ retain all selected chunk tokens, whereas the d-dimensional gists support routing. Mean, last-token, reference-end, and registered GRU pooling are configurable; detached mean pooling is the default. A

selected set $\mathcal{A}_{b,\ell}$ is materialized independently for batch row b. Local attention remains causal, while reference memory is a separate non-causal cross-attention source:

$$O_{\text{local}} = \text{softmax}((QK^\top + M_{\text{causal}})/\sqrt{d_h})V, \tag{12}$$

$$O_{\text{mem},b} = W_{O,\text{mem}} \,\text{merge}\left[\text{softmax}(Q_b K^\top_{\mathcal{A}_b,\ell}/\sqrt{d_h})V_{\mathcal{A}_b,\ell}\right], \tag{13}$$

$$O = O_{\text{local}} + \alpha O_{\text{mem}}. \tag{14}$$

The two attention operations answer different questions. Causal self-attention asks which earlier prompt tokens matter at each local position. Memory cross-attention asks which token details inside the already selected reference chunks matter. In the default `selected_chunks` mode only routed chunks enter this second computation. `full_reference` opens every chunk in a selected URI and therefore removes the chunk-selection bottleneck; `gist_only` exposes one position per selected chunk and tests what is lost when detailed token memory is withheld.

Variable memory lengths can be processed row by row, in one masked rectangle, or in deterministic length buckets. Empty rows receive an exact zero memory residual, including when the output projection has bias. The prototype can zero-initialize $W_{O,\text{mem}}$ for isolated phase-2 adaptation. This makes the memory contribution exactly zero at initialization and preserves the local path when only that projection is trained. It is a useful engineering control, not an intrinsic no-forgetting theorem for PRA.

Listing 1: Reference-memory construction and consumption in Paper 1

```
row_caches = []
for sample in batch: # private semantic address space
    cache = PRASimpleMemoryCache()
    builder = RecursiveReferenceCacheBuilder(model, resolver,
                                             tokenizer, cache, cfg)
    budget = builder.new_budget()
    for handle in sample.references:
        builder.ensure_cached(handle.uri, budget=budget)
        # ensure_cached resolves children first, partitions each object,
        # and stores per-layer token K/V plus one routing gist per chunk.
    row_caches.append(cache)

batch_cache = PRABatchedMemoryCache(row_caches)
model.set_pra_cache(batch_cache)
logits = model(tokens) # one [B,T] prompt forward
```

Listing 2: Simplified shape of routing and materialization in one PRA layer

```
class PRAttention(nn.Module):
    def forward(self, x, cache, causal_mask):
        q, k, v = self.project_local(x)
        local = causal_attention(q, k, v, causal_mask)
        q_last = merge_heads(q[:, :, -1]) # [B,d_model]
        hits_by_row = cache.route_chunks(q_last, layer=self.layer_id,
                                         top_refs=self.k_ref,
                                         top_chunks=self.k_chunk)
        rows = []
        for row, hits in enumerate(hits_by_row):
            mem_k, mem_v = cache.materialize_selected(hits,
                                                      layer=self.layer_id)
            rows.append(cross_attention(q[row], mem_k, mem_v)
                        if hits else torch.zeros_like(local[row]))
        return local + self.memory_out(torch.stack(rows))
```

The second listing deliberately omits head reshaping, row ownership arguments, rectangular masks, and diagnostics. Its point is the boundary between the small objects used to rank chunks and the full token K/V tensors used after selection. The actual batch wrapper guarantees that query row b searches only cache namespace b, even when two rows reuse the same URI string. Unequal memory lengths are padded or bucketed only after selection, with masks preventing padding or another row's content from becoming evidence. Conflating routing gists with detailed K/V would instead turn a routing approximation into a lossy memory representation.

2.5 Reference-data construction and leakage controls

The version-1 pilot split a WikiText-2 article into one-to-five URI-addressed earlier parts and a held-out continuation, but regenerated examples from each model seed and labeled every earlier part relevant. It is retained as engineering history, not causal selection evidence. Version 2 fixes data generation with seed 1729 and shares the same 512-example corpus across architectures and model seeds. It yields 409/51/52 train/validation/test examples, assigns candidate identifiers and order independently of relevance, designates one adjacent source part as the expected reference, and records source/split provenance. The candidate-count-weighted test chance rate is 0.4202.

Listing 3: Reproducible version-2 reference corpus construction

```
rng = Random(1729)
entries = parse_wikitext_articles(raw_text)
for article_id, entry in stable_order(entries):
    parts = split_natural_units(entry, min_parts=2, max_parts=6)
    target_i = choose_target_with_predecessor(parts, rng)
    positive = target_i - 1
    distractors = sample_other_prior_parts(parts, positive, rng)
    candidates = rng.shuffle([positive] + distractors)
    emit(prompt=make_handles(candidates), answer=parts[target_i],
         expected_uri=uri(article_id, positive),
         source_hash=sha256(entry), generator_seed=1729)
fixed_train, fixed_val, fixed_test = split_once(examples, 409, 51, 52)
```

Leakage controls are part of the architecture, not merely data hygiene. Candidate IDs must not encode relevance; all architectures must receive identical examples; train, validation, and test source partitions must be stable; reference content must be private to its example; and batched physical storage must preserve a sequence-to-cache mask. Content-only, random-URI, same-URI/wrong-version, shuffled, irrelevant, empty, oracle, and all-content-local variants are required to detect shortcuts. The adjacent-part label is still only a programmatic proxy; future data must mark human-verified necessary evidence.

2.6 Complexity and the systems claim

Let L be decoder layers, L_p PRA layers, T local tokens, r resolved references, m_i tokens in object i, G available routing gists, and M selected resident memory tokens. Thus G determines how many small vectors are scored, whereas M determines how much detailed evidence reaches attention. Ignoring constant head factors, the stages cost

Stage	Asymptotic work	What must be measured
resolve	$O(r)$ lookups plus source I/O	authorization, version check, bytes and latency
encode	$O(Ld\sum_i m_i^2)$	cold tokenization/encoding and accelerator transfer
cache projection	$O(L_p d^2 \sum_i m_i)$	per-layer projection, bytes written, deduplication
route	$O(Gd)$ exact gist scoring	gist count, hierarchical top-k latency, recall and calibration
cross-attend	$O(L_p TMd)$	selected tokens, layer utilization, synchronized latency
resident KV	$2wL_pMd$ bytes	allocator overhead, hit rate, eviction and tenant isolation

These terms expose the central unproven systems hypothesis. A shorter prompt is not an efficiency win if all references are re-encoded for every request. PRA becomes plausible when M is small, resolution avoids unnecessary objects, encoded KVs are reused across tokens or authorized requests, or quality improves at a fixed total cost. Results must therefore report cold, warm, and shared-cache regimes and separate lookup hits from *avoided encoding work*.

2.7 What the preliminary experiments establish—and do not establish

All models have four layers, four heads, width 256, context 128, dropout 0.1, and a 2,000-token BPE vocabulary. The initial two-seed pilot processes 524,288 tokens. The controlled follow-up uses seeds 1, 7, and 21, 1,048,576 ordinary-language tokens, fixed reference data, and 512 reference-stage optimizer steps. Parameter counts are 4,216,320 for same-width SelfAttention, 4,479,488 for full PRA, and 4,347,904 for the two-local/ two-PRA hybrid. FFN-expanded controls contain 4,478,976 and 4,347,648 parameters.

Full PRA is 0.0243 loss units below its nearly matched control, while the hybrid is 0.0323 above its matched control. At this scale, those aggregate differences motivate larger replicated comparisons; they do not establish superiority. More importantly, these ordinary-language runs test decoder compatibility rather than reference routing.

The fixed reference task used the earlier v1 summary-level router, not the corrected chunk-gist mechanism described above. In that historical system, valid-reference loss after frozen-path training is 4.7052 for full PRA and 4.7932 for hybrid PRA. Complete disabling raises loss by 0.0292 and 0.0214, but shuffled and irrelevant penalties are only 0.0064–0.0088 and oracle content changes loss by at most 0.0018. Top-1 selection remains near or below chance. The branch contributes, but correct content has not been shown to cause the gain. The v1 experiment therefore passed compatibility and preservation gates but failed its causal-selection gate. It cannot validate or invalidate the corrected router until the matched interventions are rerun against the new implementation.

Table 1: Controlled three-seed ordinary-language results transcribed from Paper 1. Population standard deviations are across seeds 1, 7, and 21.

Model	Parameters	WikiText-2 test loss
SelfAttention, same width	4,216,320	4.6378 ± 0.0380
SelfAttention, matched to full PRA	4,478,976	4.5706 ± 0.0037
Full PRA	4,479,488	4.5463 ± 0.0270
SelfAttention, matched to hybrid	4,347,648	4.5725 ± 0.0108
Hybrid PRA	4,347,904	4.6049 ± 0.0359

The source manuscript reports aggregate seed means and population standard deviations, not a complete seed-by-seed ledger for every run. A submission version must export one row per run (architecture, seed, data hash, checkpoint hash, trainable parameters, tokens, loss, uncertainty inputs, timing, and counterfactual condition) rather than attempting to reconstruct individual values from aggregates.

2.8 Counterfactual doctrine and falsifiable comparison

For loss L, define

$$\Delta_{\text{use}} = L_{\text{disabled}} - L_{\text{valid}}, \qquad \Delta_{\text{content}} = L_{\text{shuffled}} - L_{\text{valid}}. \tag{15}$$

$\Delta_{\text{use}} > 0$ only shows that activating the path helps. A credible content claim requires a reproducible $\Delta_{\text{content}} > 0$, a similar penalty for irrelevant references, increasing advantage on locally insufficient examples, and routing above a candidate-count-aware chance baseline. Attention weights are diagnostics; removal, substitution, stale-version, and mediation-style interventions supply stronger evidence.

The decisive experiment crosses selection and transport: oracle versus learned selection; prompt-text, cross-attention, and native-KV transport; valid/disabled/shuffled/irrelevant/ oracle content; and cold/warm/shared caches. It matches model parameters, trainable parameters, source corpus, visible and retrieved token budgets, optimizer tokens, FLOPs, peak memory, and synchronized latency. PRA's efficiency thesis is falsified in a target regime if it does not improve the quality–cost frontier after reference encoding and cache misses are included, or if correct content does not outperform corrupted controls.

PRA has now supplied a concrete example of the survey's five-part vocabulary: handles name objects, gists support selection, the resolver binds names to versioned content, the cache materializes layer-specific token K/V, and cross-attention fuses selected evidence with the local stream. The remaining sections widen the comparison. Some methods change the local interaction graph, some preserve state over time, and others move selection outside the model entirely. Reading them through the same five operations makes both overlaps and architectural differences easier to identify. The next seven sections apply the access-contract vocabulary to representative systems. To keep the mathematics comparable, T denotes sequence length, d model width, and d_h the width of one attention head. The matrices Q, K, and V contain projected queries, keys, and values; M_{causal} masks positions that a causal decoder may not read. Method-specific symbols are introduced where they appear. Complexity expressions describe the dominant conceptual operation and omit constant head and implementation factors unless those factors are themselves the subject of the method.

3 Category I: Dense, Sparse, and Approximate Long-Context Attention

This family changes the token-interaction graph after information has already been materialized as one sequence. It is therefore the cleanest "all evidence local" control for PRA: selection is either absent or encoded in the attention mask, object identity is reduced to position, and no cross-request materialization is implied.

3.1 Dense attention and positional extension

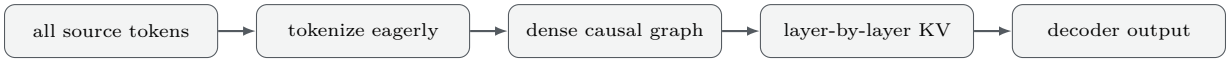
Motivation and mechanism. Dense causal attention is the least structured but most general access rule: every position can read every earlier position. RoPE scaling, position interpolation, and long-context continued pretraining extend the usable position range without changing this graph. For T tokens,

$$A = \text{softmax}((QK^\top + M_{\text{causal}})/\sqrt{d_h})V, \qquad \text{time } O(T^2 d), \text{ attention storage } O(T^2). \tag{16}$$

Positional extrapolation and attention efficiency are different problems. A model may accept a long index while still allocating quadratic work, or execute efficiently while failing to use distant positions.

Listing 4: Dense causal attention; PyTorch uses optimized kernels when possible

```
def dense_causal(q, k, v): # [B,H,T,D]
    return F.scaled_dot_product_attention(q, k, v, is_causal=True)
```



Evidence, failure modes, and lineage. Standard language modeling and long-context adaptations use perplexity, long-document QA, retrieval probes, and throughput/memory sweeps. Dense attention is a strong baseline when sources are short or densely relevant, and modern fused kernels make it unexpectedly competitive at moderate lengths. Its failures include quadratic cost, distraction, positional degradation, and lack of stable object identity or reuse beyond exact prefixes. Needle tests diagnose reachability but do not establish synthesis or citation behavior.

Relation and comparison to PRA. Dense context performs eager materialization; PRA advertises a potentially larger logical context and materializes selected objects. Compare them on the same source corpus under (i) equal visible-plus-retrieved tokens, (ii) equal FLOPs, (iii) equal peak KV bytes, and (iv) equal synchronized latency. Sweep evidence density and object reuse. Dense attention should be expected to win when most tokens matter once; PRA's hypothesis is strongest when few named objects serve many queries. The all-content-local condition is also PRA's transport upper control.

3.2 Longformer

Motivation and mechanism. Longformer observes that document dependencies are mostly local but a few task positions need global connectivity. A token attends within a window and, optionally, to a set $\mathcal{G}$ of global positions [Beltagy et al., 2020]:

$$\mathcal{N}(i) = \{j : |i - j| \le w/2\} \cup \mathcal{G}, \qquad O(Tw + T|\mathcal{G}|). \tag{17}$$

Local layers propagate information gradually; global tokens create short paths across the document and can be task-selected (for example, question tokens in QA). Unlike truncation, the model retains an explicit sparse graph over the entire input.

Listing 5: Educational Longformer mask; production kernels avoid materializing T^2

```
def longformer_mask(T, window, global_pos, device):
    i = torch.arange(T, device=device)
    local = (i[:, None] - i[None, :]).abs() <= window // 2
    g = torch.zeros(T, dtype=torch.bool, device=device)
    g[global_pos] = True
    return local | g[:, None] | g[None, :]
```

Evidence and paper trail. The original paper studies character language modeling, long-document classification, and QA; its encoder-decoder continuation LED applies the pattern to long-input generation. Comparisons include RoBERTa-style dense baselines and task-specific long-document systems, using bits-per-character, accuracy/F1, and generation metrics. Longformer became a template for domain adaptations and local/global attention. Its main failure is structural: useful long-range routes must be known through the global mask or emerge through multiple local hops. Fixed windows can split tables, citations, or cross-section evidence, and global positions can become a bottleneck.

PRA relation and matched experiment. Longformer makes relevance a sparse edge decision over already materialized tokens; PRA makes it an object-admission decision before latent transport. Give both the same documents and vary evidence sparsity, distance, and whether section boundaries are known. Match the number of attended key positions: Longformer local/global edges versus PRA top-k object tokens. Add shuffled global markers and shuffled URI content. This separates sensitivity to graph placement from sensitivity to object semantics. A hybrid can use Longformer locally inside a large resolved object and PRA between objects.

3.3 BigBird

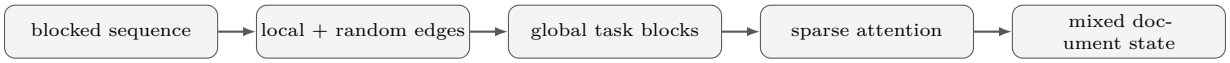
Motivation and mechanism. BigBird combines window, random, and global edges so that each token has bounded degree while the graph mixes rapidly [Zaheer et al., 2020]:

$$\mathcal{N}(i) = \mathcal{N}_{\text{window}}(i) \cup \mathcal{N}_{\text{random}}(i) \cup \mathcal{G}. \tag{18}$$

The random component supplies shortcuts without content-dependent routing; global tokens connect task anchors to all positions. Under the paper's graph assumptions this pattern retains strong expressivity results while reducing sequence complexity to linear in T for fixed block sizes.

Listing 6: Block-level BigBird adjacency sketch

```
def bigbird_blocks(n_blocks, radius, random_k, global_blocks=(0,)):
    edges = []
    for i in range(n_blocks):
        local = range(max(0, i-radius), min(n_blocks, i+radius+1))
        random = torch.randperm(n_blocks)[:random_k].tolist()
        edges.append(sorted(set(local) | set(random) | set(global_blocks)))
    return edges
```



Evidence, ablations, and limitations. BigBird evaluates natural-language QA and summarization as well as genomic sequences, comparing dense BERT/RoBERTa-style models and other efficient Transformers with task accuracy/F1, ROUGE, and sequence-scale behavior. The meaningful ablation is the contribution of local, random, and global edges. Random connectivity is inexpensive on average but can miss a particular necessary route; global tokens require a placement rule, and sparse kernels do not automatically make an otherwise identical pretrained model work at new lengths.

PRA relation and experiment. BigBird constructs content-independent shortcuts inside one anonymous token graph. PRA constructs content-dependent edges to explicitly named external objects. Compare a matched sparse-edge budget in which random/global block edges and top-k URI objects expose the same number of keys. Sweep distractor blocks, wrong handles, evidence distance, and repeated queries. Report cold and warm materialization for PRA; otherwise BigBird is unfairly charged only for attention while PRA hides encoding. An instructive hybrid replaces random inter-document edges with URI-restricted routing and retains BigBird within each selected document.

3.4 Performer

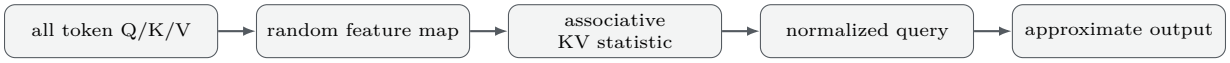
Motivation and mechanism. Performer approximates softmax attention with positive random features (FAVOR+), exchanging a $T \times T$ matrix for associative summaries [Choromanski et al., 2021]. If $\exp(q^\top k) \approx \phi(q)^\top \phi(k)$, then

$$\mathrm{Attn}(Q,K,V) \approx D^{-1}\phi(Q)\big(\phi(K)^\top V\big), \quad D = \mathrm{diag}(\phi(Q)(\phi(K)^\top \mathbf{1})). \tag{19}$$

For fixed feature count, time and memory are linear in sequence length. Causal execution uses prefix sums of $\phi(k_i)v_i^\top$ and $\phi(k_i)$.

Listing 7: Causal kernel-attention recurrence; phi must be a positive feature map

```
def causal_linear_attention(qf, kf, v, eps=1e-6):
    kv = torch.einsum('...tm,...td->...tmd', kf, v).cumsum(dim=-3)
    kz = kf.cumsum(dim=-2)
    num = torch.einsum('...tm,...tmd->...td', qf, kv)
    den = torch.einsum('...tm,...tm->...t', qf, kz).unsqueeze(-1)
    return num / den.clamp_min(eps)
```



Evidence, failure modes, and lineage. Performer reports image, text, and protein experiments and measures both downstream quality and approximation/efficiency. Its paper trail lies in kernelized linear attention and random-feature estimation. Feature variance, finite-precision instability, causal implementation details, and inability to recover a specific token after destructive aggregation are central failure modes. Linear cost does not mean selective relevance: every token contributes to the statistic.

PRA relation and comparison. Performer approximates computation after eager materialization; PRA changes which objects enter computation. The two are complementary: use linear attention for the local stream or within a resolved memory. A fair experiment crosses exact versus FAVOR+ local attention with all-local versus PRA access, reports approximation error and exact-value recall, and sweeps random-feature count under fixed FLOPs. PRA corruption tests remain necessary because a low approximation error says nothing about whether the correct reference was selected.

4 Category II: Recurrence and Compression

Recurrence changes the temporal boundary: earlier segments leave behind state rather than remaining as raw local tokens. The central distinction from PRA is *write-time retention versus read-time dereferencing*. Recurrent systems decide what history survives as it passes; PRA assumes an addressable backing object can be fetched later.

4.1 Transformer-XL

Motivation and mechanism. Transformer-XL addresses fixed context and segment fragmentation by carrying the previous segment's hidden states forward as stop-gradient memory and using relative positional attention [Dai et al., 2019]:

$$\widetilde{H}_s^{(\ell-1)} = [\mathrm{SG}(H_{s-1}^{(\ell-1)}); H_s^{(\ell-1)}], \quad Q = H_s W_Q, \quad K, V = \widetilde{H}_s W_{K,V}. \tag{20}$$

Only current-segment states receive gradients through the recurrent boundary. The memory extends effective context without backpropagating across the entire history; relative positions prevent a cached state from being confused with a current absolute position.

Listing 8: Segment recurrence; detach is the defining optimization boundary

```
def transformer_xl_segment(layer, x, old_memory, memory_len):
    kv_source = torch.cat([old_memory.detach(), x], dim=1)
    y = layer(query=x, key=kv_source, value=kv_source, relative_positions=True)
    new_memory = torch.cat([old_memory, x], dim=1)[:, -memory_len:].detach()
    return y, new_memory
```

Evidence and limitations. The paper evaluates WikiText-103, One Billion Word, Penn Treebank, enwiki8, and text8 using perplexity or bits-per-character, alongside recurrent and Transformer language-model baselines. It reports then-leading benchmark results and substantially faster evaluation than recomputing overlapping windows. Ablations isolate recurrence and relative positions. Memory is chronological, anonymous, fixed-length, and not revisable: evicted information is gone, irrelevant history consumes capacity, and a detached boundary limits credit assignment.

PRA relation and direct test. Transformer-XL answers "what recent state should remain?"; PRA answers "which named remote object should be loaded now?" Compare them on streams with either recency-correlated evidence or remote named evidence. Match resident KV bytes and total encoding FLOPs, sweep distance beyond the recurrent window, and include repeated access to the same old object. A hybrid keeps Transformer-XL memory for recent discourse while letting a URI handle reload evicted or non-contiguous source material.

4.2 Compressive Transformer

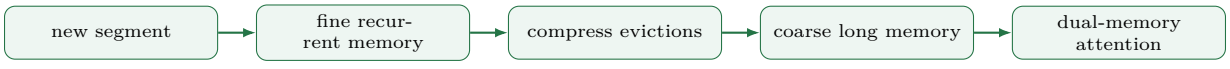
Motivation and mechanism. Compressive Transformer extends Transformer-XL with a second, lower-resolution memory. States evicted from fine memory M^m are transformed by a compressor C before entering compressed memory M^c [Rae et al., 2020]:

$$M_s^c = \mathrm{tail}_{n_c}([M_{s-1}^c; C(E_{s-1})]), \tag{21}$$

where E_{s-1} denotes the fine states being evicted. Attention reads both stores. Pooling, convolution, and usage-based compression trade temporal resolution for longer reach; auxiliary reconstruction losses encourage retained predictive information.

Listing 9: Two-tier chronological memory update

```
def update_memory(fine, compressed, new_states, fine_cap, comp_cap, compress):
    merged = torch.cat([fine, new_states], dim=1)
    evicted, fine = merged[:, :-fine_cap], merged[:, -fine_cap:]
    compressed = torch.cat([compressed, compress(evicted)], dim=1)
    return fine.detach(), compressed[:, -comp_cap:].detach()
```



Evidence and failure modes. The original work studies long-range language modeling, compares compression functions and auxiliary objectives, and uses perplexity plus attention and memory diagnostics. The method's honest trade-off is rate versus fidelity. Pooling can erase identifiers, superposition can merge distractors, learned compression can optimize average prediction while losing rare facts, and chronological stores cannot directly invalidate a stale semantic object.

PRA relation and experiment. PRA ordinarily reloads an original or cached object; Compressive Transformer obligatorily summarizes passing history. A useful hybrid attaches several materializations to one URI: metadata, summary, compressed latent, and full KV, with uncertainty-driven escalation. Compare fine-only recurrence, dual-memory compression, summary-only PRA, and full-object PRA under equal resident bytes. Sweep compression ratio, rare exact facts, semantic synthesis, and stale-object replacement. Report whether escalation recovers failures and charge full-object encoding in cold-cache trials.

4.3 Recurrent Memory Transformer (RMT)

Motivation and mechanism. RMT inserts a small set of learned memory tokens into every segment. Their output states become the memory-token inputs to the next segment, creating an end-to-end learned bottleneck [Bulatov et al., 2022]. If m_s are incoming memory tokens and x_s current tokens,

$$[\hat{m}_s, H_s, m_{s+1}] = \text{Transformer}([m_s, x_s, m_s^{\text{write}}]). \tag{22}$$

The exact placement varies by implementation, but the conceptual operation is stable: read old slots, process the segment, and write a fixed number of new slots.

Listing 10: RMT segment interface

```
def rmt_step(model, tokens, memory, write_tokens):
    sequence = torch.cat([memory, tokens, write_tokens], dim=1)
    hidden = model(sequence)
    next_memory = hidden[:, -write_tokens.size(1):]
    token_hidden = hidden[:, memory.size(1):-write_tokens.size(1)]
    return token_hidden, next_memory
```

Evidence, lineage, and failures. RMT is evaluated on language modeling and algorithmic/long-sequence tasks; later work demonstrates extrapolation on very long synthetic problems. Baselines include finite-window Transformers and recurrent-memory variants, with loss, task accuracy, and length extrapolation as metrics. The learned bottleneck is simple and differentiable, but it must anticipate future needs at write time. Rare details can be overwritten, slot semantics lack stable provenance, and backpropagation depth or memory curricula strongly affect training.

PRA relation and experiment. RMT learns *what to remember while reading*; PRA learns *what to fetch while answering*. Compare both on matched memory bytes under two datasets: one where future questions are predictable during ingestion and one where they are not. Substitute or reset RMT slots and shuffle PRA references to measure content use. A hybrid stores an RMT summary in the URI metadata tier but preserves the source object for read-time expansion when the summary is insufficient.

4.4 Block-Recurrent Transformer

Motivation and mechanism. Token-wise recurrence is sequential, whereas ordinary self-attention is parallel but bounded by its active window. Block-Recurrent Transformer places the recurrent boundary at a block of tokens. A set of state vectors S_{b-1} and the current token block X_b exchange information through self- and cross-attention, then LSTM-style gates form the state passed to the next block [Hutchins et al., 2022]:

$$\widetilde{X}_b = \text{TokenBlock}(X_b, S_{b-1}), \tag{23}$$

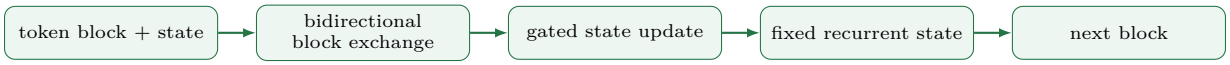
$$\widetilde{S}_b = \text{StateBlock}(S_{b-1}, X_b), \tag{24}$$

$$S_b = g_b \odot S_{b-1} + (1 - g_b) \odot \widetilde{S}_b. \tag{25}$$

The exact cell contains several gates, but the computational principle is visible here: parallel token processing occurs inside a block, while a fixed set of latent states carries information between blocks. For fixed block and state sizes, total work grows linearly with the number of processed tokens.

Listing 11: Simplified block-recurrent update

```
def block_recurrent_step(token_block, state, cell):
    token_out = cell.tokens_attend(token_block, state)
    state_candidate = cell.state_attends(state, token_block)
    gate = torch.sigmoid(cell.gate(torch.cat([state, state_candidate], -1)))
    next_state = gate * state + (1 - gate) * state_candidate
    return token_out, next_state
```



Evidence and limitations. The original study evaluates PG19 books, arXiv papers, and GitHub source code against long-range Transformer-XL variants, reporting language-model loss and execution behavior. The design uses accelerator-friendly block parallelism, but the state remains an anonymous write-time bottleneck. A gate can preserve a feature without preserving its source, version, or exact wording, and later queries cannot reload discarded detail.

PRA relation and experiment. Block recurrence and PRA solve different halves of the working-set problem. The recurrent state keeps a compact summary of the path already read; PRA can dereference a named object that was never read or whose details no longer fit the state. Compare recurrent state, PRA, and a hybrid under matched resident bytes on tasks that independently vary temporal continuity and source addressability. State resets test dependence on recurrent content; shuffled URI objects test dependence on PRA content.

4.5 TransformerFAM

Motivation and mechanism. A sliding-window transformer eventually loses access to every old token, and a detached segment cache limits how gradients teach long-range retention. TransformerFAM introduces a short Feedback Attention Memory (FAM) whose current activations attend with each token block and whose updated activations feed the next block [Hwang et al., 2024]. If F_{b-1} is the previous feedback state, one abstract update is

$$[H_b, F_b] = \mathrm{MaskedAttn}([X_b, F_{b-1}], [X_b, M_b, F_{b-1}]), \tag{26}$$

where M_b denotes the ordinary block-sliding memory. The feedback queries are contextual: they change after every block instead of using one static summary query. FAM adds no new projection weights when it reuses the transformer's attention parameters, although it changes training, masks, positions, and recurrent execution.

Listing 12: Feedback attention memory across blocks

```
def fam_step(block, sliding_memory, fam, attention, mask):
    queries = torch.cat([block, fam], dim=1)
    keys_values = torch.cat([block, sliding_memory, fam], dim=1)
    outputs = attention(queries, keys_values, keys_values, attn_mask=mask)
    return outputs[:, :block.size(1)], outputs[:, block.size(1):]
```

Evidence and limitations. The paper studies models at 1B, 8B, and 24B parameters on language modeling, passkey retrieval, NarrativeQA, and related long-context tasks. Its ablations show that static summary queries and some simpler feedback variants do not retain the passkey as well as contextual FAM. Improvements over block sliding attention can be modest on several natural tasks, and the latent slots still lack stable object identity, selective invalidation, or exact reconstruction after compression.

PRA relation and experiment. FAM is dynamic working memory; PRA is addressed backing memory. A useful hybrid lets FAM carry the current reasoning state while PRA supplies source detail. Compare a static summary token, contextual FAM, content-gist PRA, and the hybrid. Use delayed exact recall, multi-hop synthesis, stale-source replacement, and unseen question types. The key question is whether a small feedback state reduces PRA faults without hiding content failures inside an uninterpretable latent bottleneck.

4.6 Infini-attention

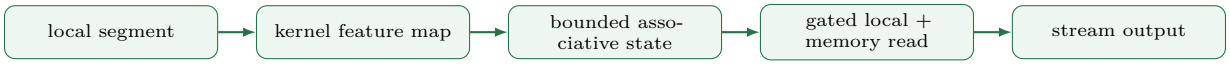
Motivation and mechanism. Infini-attention combines masked local attention with a bounded associative memory inside each block [Munkhdalai et al., 2024]. With positive feature map ϕ, a segment updates

$$M_s = M_{s-1} + \phi(K_s)^\top V_s, \qquad z_s = z_{s-1} + \sum_i \phi(k_{s,i}), \tag{27}$$

and reads $\phi(Q_s)M_{s-1}/(\phi(Q_s)z_{s-1})$. A learned gate mixes this compressive read with dot-product attention over the current segment. Memory size is independent of total processed length, which makes streaming execution possible.

Listing 13: Associative-memory read and write

```
def infini(qf, kf, v, M, z, eps=1e-6):
    read = torch.einsum('btd,bdh->bth', qf, M)
    denom = torch.einsum('btd,bd->bt', qf, z).unsqueeze(-1)
    read = read / denom.clamp_min(eps)
    M = M + torch.einsum('btd,bth->bdh', kf, v)
    z = z + kf.sum(dim=1)
    return read, M.detach(), z.detach()
```



Evidence and limitations. The paper evaluates long-context language modeling, one-million-token passkey retrieval, and 500K-token book summarization with 1B- and 8B-scale models. Metrics include perplexity, retrieval accuracy, summarization quality, and memory scaling. Bounded state is the strength and the limitation: many keys superpose, collisions and normalization can obscure exact facts, and there is no explicit object version, authorization boundary, or selective invalidation.

PRA relation and direct comparison. Infini-attention offers constant-size lossy history; PRA offers selective identity with potentially unbounded backing storage and bounded resident working set. Match resident bytes and local window, then sweep number of objects, repeated keys, exact-value queries, semantic summaries, and stale replacements. PRA must include resolution/encoding latency. A hybrid maintains cheap Infini state for the stream while a collision, low-confidence query, or explicit handle faults in a named PRA object.

4.7 Titans and Test-Time Neural Memory

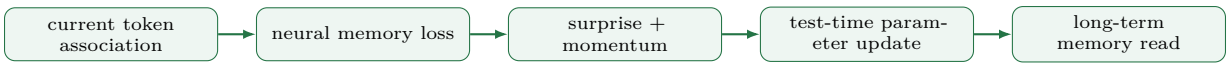
Motivation and mechanism. Fixed-state recurrent models update activations, but their update rule is usually shallow. Titans instead treats a neural network $\mathcal{M}_{\theta_t}$ as long-term memory and updates its parameters while processing the test sequence [Behrouz et al., 2025]. For a key–value association (k_t, v_t), an associative loss measures how poorly the current memory predicts the value,

$$\ell_t(\theta_t) = \|\mathcal{M}_{\theta_t}(k_t) - v_t\|_2^2, \qquad s_t = \|\nabla_{\theta_t}\ell_t\|, \tag{28}$$

and the gradient magnitude s_t acts as a surprise signal. Momentum and forgetting terms control how the parameter update incorporates surprising events. Titans combines this neural long-term memory with precise short-window attention and data-independent persistent memory. Its Memory-as-Context, Memory-as-Gate, and Memory-as-Layer variants differ in where the long-term read enters the sequence model.

Listing 14: Conceptual surprise-weighted test-time memory update

```
def neural_memory_step(memory, key, value, optimizer_state):
    prediction = memory(key)
    loss = F.mse_loss(prediction, value)
    gradients = torch.autograd.grad(loss, memory.parameters())
    surprise = global_norm(gradients)
    optimizer_state = momentum_and_forgetting(optimizer_state, gradients, surprise)
    apply_test_time_update(memory, optimizer_state)
    return prediction, surprise
```



Evidence and limitations. Titans reports language modeling, common-sense reasoning, genomics, time-series, and long-context retrieval experiments, including context lengths beyond two million tokens. Ablations examine memory depth, momentum, forgetting, convolution, persistent memory, and the three integration variants. These results motivate test-time neural memory, but online parameter updates introduce sequential optimization, state-reset, reproducibility, privacy, and contamination questions. The memory is content addressed rather than an authoritative source store; forgetting or correcting one object is not a simple URI invalidation.

PRA relation and experiment. Titans learns a compressed latent history online; PRA resolves versioned external objects and transports selected evidence. Compare them on four operations: repeated semantic patterns, exact quotations, correction of stale facts, and access to an object never seen in the current stream. Cross both methods with state reset, corrupted evidence, and repeated-query conditions. A hybrid could use Titans for experience-dependent routing priors while PRA retains inspectable source identity. Such a hybrid must report whether test-time updates improve object selection or merely adapt the decoder independently of supplied content.

5 Category III: Differentiable and Episodic External Memory

These methods move state outside the ordinary recurrent or token representation while keeping access near neural computation. They introduce two questions that recur in PRA: who writes the memory, and whether an address denotes a stable object or only a similar vector.

5.1 Memory Networks

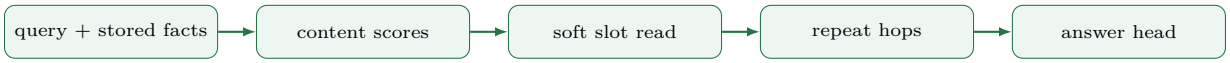
Motivation and mechanism. Memory Networks separate input encoding, memory storage, addressing, and response production [Weston et al., 2015]. Facts become slots m_i; a query q produces a content read

$$p_i = \mathrm{softmax}_i(q^\top m_i), \qquad o = \sum_i p_i c_i, \qquad q' = q + o. \tag{29}$$

Repeating the read implements multi-hop reasoning. End-to-end variants learn embeddings from question-answer supervision; earlier variants used stronger supporting-fact labels.

Listing 15: One differentiable memory hop

```
def memory_hop(query, address_slots, content_slots):
    weights = torch.softmax(query @ address_slots.transpose(-1, -2), dim=-1)
    read = weights @ content_slots
    return query + read, weights
```



Evidence, lineage, and failure modes. The paper trail runs from supervised Memory Networks through end-to-end memory networks and key-value memory networks. Evaluations include synthetic bAbI reasoning and question answering, using answer accuracy and supporting-fact retrieval. Multiple hops make compositional access inspectable, but soft attention across a large flat store scales poorly; embeddings can exploit lexical shortcuts; and overwrite, provenance, and slot lifecycle are under-specified for production knowledge.

PRA relation and experiment. Memory Networks establish the learned-read primitive that PRA reuses, but PRA delegates object writes and identity to a resolver/cache. Compare flat learned slots with URI candidate restriction followed by the same within-candidate attention. Hold stored facts and read hops fixed; sweep memory size, paraphrase, distractors, and wrong-object substitutions. Report supporting-fact recall and counterfactual answer change, not attention maps alone. The expected trade-off is open semantic discovery versus cheap exact restriction when a handle is available.

5.2 Differentiable Neural Computer (DNC)

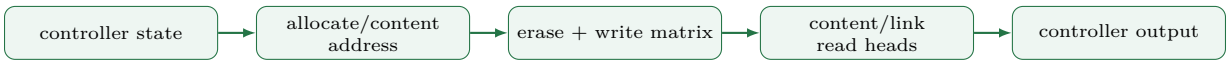
Motivation and mechanism. The DNC couples a neural controller to a read/write matrix with content lookup, temporal links, usage-based allocation, and differentiable erase/add operations [Graves et al., 2016]. A write updates row i as

$$M_t(i) = M_{t-1}(i) \odot (1 - w_t^w(i)e_t) + w_t^w(i)a_t, \tag{30}$$

while read weights mix content addressing with forward/backward traversal of a learned temporal linkage. It therefore supports both associative recall and ordered walks.

Listing 16: Core DNC erase/add write; allocation and link updates omitted

```
def dnc_write(memory, write_weight, erase, add):
    w = write_weight.unsqueeze(-1)
    return memory * (1 - w * erase.unsqueeze(1)) + w * add.unsqueeze(1)
```



Evidence and limitations. DNC experiments emphasize synthetic graph traversal, question answering, and algorithmic tasks, comparing recurrent controllers and earlier memory systems with exact task accuracy or sequence error. The machine can learn compact algorithm-like behavior, yet many interacting gates make training brittle. Soft access is expensive at large slot counts, learned addresses lack stable external meaning, and catastrophic write errors can corrupt later computation.

PRA relation and experiment. DNC learns both memory management and access; PRA externalizes durable writes, versions, and permissions, narrowing learning to selection and fusion. Compare DNC, read-only DNC, and PRA on a graph task whose nodes have stable IDs plus a variant with IDs hidden. Match active memory bytes and controller parameters. Test node substitution, temporal traversal, stale updates, and multi-tenant isolation. A hybrid can use a small DNC as transient scratch space while PRA supplies immutable, auditable backing objects.

5.3 Memformer

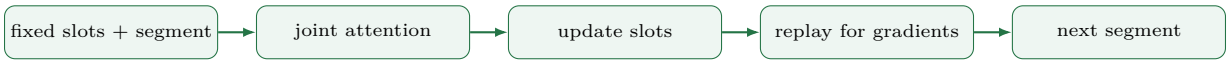
Motivation and mechanism. Memformer carries a fixed number of dynamic memory slots across segments and introduces memory replay backpropagation so training need not retain every old activation [Wu et al., 2020]. A segment attends to slots and local tokens, then a write operation updates the slots:

$$S_{t+1} = \mathrm{Write}(S_t, H_t), \qquad |S_t| = N_s \text{ constant.} \tag{31}$$

Replay reconstructs selected memory states during backward computation, trading extra compute for lower activation storage.

Listing 17: Simplified fixed-slot segment update

```
def memformer_step(block, tokens, slots):
    joint = torch.cat([slots, tokens], dim=1)
    hidden = block(joint)
    new_slots = hidden[:, :slots.size(1)]
    token_hidden = hidden[:, slots.size(1):]
    return token_hidden, new_slots
```



Evidence, ablations, and failures. The paper evaluates sequence modeling under long inputs, reporting task quality, training memory, and speed against Transformer and memory-augmented baselines. Slot count, recurrence, and replay are key ablations. Constant space is attractive, but slots are lossy summaries without stable semantic names; useful facts may be overwritten; replay adds complexity; and a slot cannot be independently revoked or refreshed when its source changes.

PRA relation and experiment. Memformer provides a learned working set; PRA adds a named backing store and explicit faults. Use identical segment encoders and match resident slots/KV bytes. Ask questions whose evidence is either predictable during ingestion or selected only after ingestion, and sweep time-to-query. Compare slot reset/substitution with PRA disabled/shuffled controls. A tiered design stores Memformer summaries as cheap URI metadata and faults in full layer-specific memory when summary confidence is low.

5.4 Memorizing Transformer

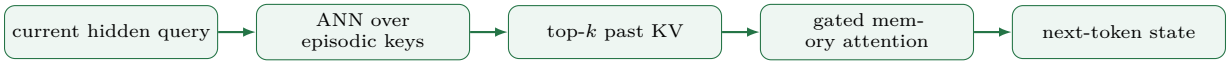
Motivation and mechanism. Memorizing Transformer records past key/value pairs in a large non-differentiable episodic database and retrieves nearest keys for current queries [Wu et al., 2022]. For query q,

$$\mathcal{I} = \mathrm{kNN}(q, \{k_i\}), \qquad o_{\mathrm{mem}} = \mathrm{softmax}(qK_{\mathcal{I}}^{\top})V_{\mathcal{I}}, \tag{32}$$

which is gated with ordinary attention. The store can span many batches and does not need gradients through nearest-neighbor selection.

Listing 18: Episodic KV retrieval; an ANN index replaces the dense distance

```
def memorizing_read(q, key_db, value_db, k):
    dist = torch.cdist(q.float(), key_db.float())
    idx = dist.topk(k, largest=False).indices
    keys, values = key_db[idx], value_db[idx]
    score = torch.einsum('...d,...kd->...k', q, keys)
    return torch.einsum('...k,...kd->...d', score.softmax(-1), values)
```



Evidence and failure modes. Experiments span language-model datasets and model scales up to billions of parameters, measuring perplexity versus memory size and retrieval configuration. The method demonstrates that latent episodic retrieval can improve a Transformer without differentiating through a huge store. Its risks are near-duplicate memorization, index cost, stale states after model updates, cross-example contamination, and weak source provenance. Similarity can retrieve the right surface continuation while failing compositional evidence use.

PRA relation and experiment. This is a close latent-memory precursor. The main difference is open kNN over anonymous stream states versus two-stage object identity and within-object attention. Compare (a) global episodic kNN, (b) exact URI restriction, (c) semantic object routing then token kNN, and (d) oracle object restriction. Hold total indexed KVs and returned keys fixed. Include near-duplicate continuation, multi-object composition, wrong URI, and stale-model cache conditions, and report source-level recall alongside perplexity.

6 Category IV: Retrieval-Augmented Pretraining and Generation

Retrieval augmentation places a corpus outside model parameters. The decisive axes are whether selection is supervised or latent, whether retrieved material enters as text or hidden state, and whether retrieval occurs once, per chunk, or per generated token.

6.1 kNN-LM

Motivation and mechanism. kNN-LM stores training-context representations paired with observed next tokens. At inference it builds a distance-weighted neighbor distribution and interpolates it with the parametric language model [Khandelwal et al., 2020]:

$$p_{\mathrm{kNN}}(y \mid q) \propto \sum_{i:y_i=y} \exp(-d(q,k_i)/\tau), \tag{33}$$

$$p(y \mid q) = \lambda p_{\mathrm{kNN}}(y \mid q) + (1-\lambda) p_{\mathrm{LM}}(y \mid q). \tag{34}$$

No retrieved passage is read compositionally; memory changes the final token distribution.

Listing 19: kNN-LM probability interpolation

```
def knn_lm(log_p_lm, q, keys, next_tokens, k, vocab, lam, tau):
    d, idx = torch.cdist(q, keys).topk(k, largest=False)
    weights = torch.softmax(-d / tau, dim=-1)
    p_knn = torch.zeros(*q.shape[:-1], vocab, device=q.device)
    p_knn.scatter_add_(-1, next_tokens[idx], weights)
    return torch.logaddexp(log_p_lm + math.log1p(-lam),
                           p_knn.clamp_min(1e-12).log() + math.log(lam))
```

Evidence, lineage, and limitations. The original work reports perplexity on WikiText-103 and domain-adaptation settings, comparing the base LM and cache/retrieval baselines. Improvements without parameter updates motivate later datastore adaptation, pruning, and efficient kNN work. Storage and ANN latency can be large; interpolation needs calibration; privacy risks follow from training-example memorization; and the mechanism is strongest for continuation similarity rather than multi-fact reasoning.

PRA relation and comparison. kNN-LM fuses at the output distribution; PRA fuses rich memory within hidden computation. Compare them with the same corpus encoder and retrieval budget on near-duplicate prediction versus questions requiring composition. Report perplexity, calibration, exact-match, datastore bytes, and latency. Add shuffled token labels for kNN-LM and shuffled objects for PRA. If PRA only matches kNN-LM on near-copy tasks at higher cost, its richer transport is not justified.

6.2 REALM

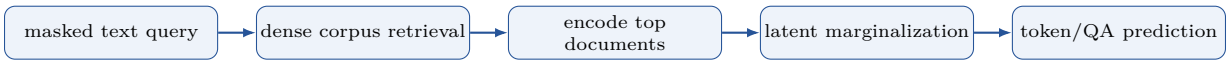
Motivation and mechanism. REALM jointly pretrains a dense retriever and a masked language model over an external corpus [Guu et al., 2020]. With latent document z,

$$p(y \mid x) = \sum_{z \in \mathrm{TopK}(x)} p_\eta(z \mid x) p_\theta(y \mid x, z), \tag{35}$$

where approximate maximum inner-product search supplies candidates and the language loss provides a learning signal to the retriever. Retrieved text is encoded jointly with the masked input, so the knowledge store can change without retraining all model parameters.

Listing 20: REALM-style latent-document marginalization

```
scores, doc_ids = retriever.topk(query_encoder(masked_input), k)
log_p_doc = scores.log_softmax(-1)
log_p_y = torch.stack([reader(masked_input, corpus[i]) for i in doc_ids], -1)
loss = -torch.logsumexp(log_p_doc + log_p_y, dim=-1).mean()
```



Evidence and failure modes. REALM evaluates open-domain QA and knowledge-intensive pretraining, comparing retrieval-free language models and retrieval pipelines with exact match and retrieval diagnostics. Its important legacy is end-to-end retriever learning from the task objective. Index refresh is expensive, top-k truncation biases the latent sum, corpus snapshots become stale, and retrieval may learn shortcuts when answers are predictable locally.

PRA relation and experiment. PRA should borrow REALM's trainable selection rather than rely on fixed summary cosine scores. REALM materializes retrieved text in an encoder; PRA can reuse per-object layer KVs and explicit handles. Cross oracle/learned retrieval with text/cross-attention transport on the same corpus. Match retrieved tokens and index queries; evaluate locally sufficient and reference-required subsets, shuffled documents, random handles, warm caches, and index refresh after source updates.

6.3 Retrieval-Augmented Generation (RAG)

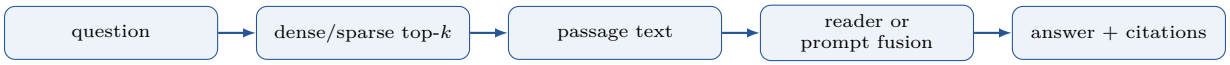
Motivation and mechanism. RAG combines a dense retriever with a seq2seq generator. RAG-Sequence holds a document mixture for a whole output; RAG-Token permits the latent document to vary by token [Lewis et al., 2020]:

$$p(y \mid x) = \prod_t \sum_{z \in \mathrm{TopK}(x)} p_\eta(z \mid x) p_\theta(y_t \mid x, z, y_{<t}). \tag{36}$$

In common systems practice, retrieved passages are concatenated into a prompt rather than exactly marginalized, but the selection/materialization distinction is the same.

Listing 21: Minimal retrieve-then-read RAG pipeline

```
def rag_answer(question, retriever, generator, k):
    hits = retriever.search(question, k=k)
    context = "\n\n".join(f"[{h.id}] {h.text}" for h in hits)
    return generator(f"Sources:\n{context}\nQuestion: {question}\nAnswer:")
```



Evidence, descendants, and failures. RAG evaluates Natural Questions, TriviaQA, WebQuestions, CuratedTREC, MS MARCO-style knowledge tasks, and generation, comparing DPR, closed-book, and retrieval-augmented baselines with exact match and generation/diversity metrics. It seeded extensive work on reranking, query rewriting, corrective and agentic RAG. Failure modes include chunk boundary errors, retrieval miss, distractor sensitivity, prompt-token cost, citation mismatch, and stale indexes. A fluent answer does not prove that retrieved evidence caused the output.

PRA relation and direct comparison. Both keep knowledge external; they differ in timing and materialization. RAG usually chooses text before generation, while PRA can expose named latent objects by layer or step. Use one corpus, retriever, generator backbone, and evidence budget. Compare prompt text, encoder cross-attention, cached object KVs, and iterative retrieval. Report task quality, citation precision/recall, valid/disabled/shuffled effects, cold/warm latency, tokenization/encoding cost, and behavior as queries over the same documents repeat.

6.4 Atlas

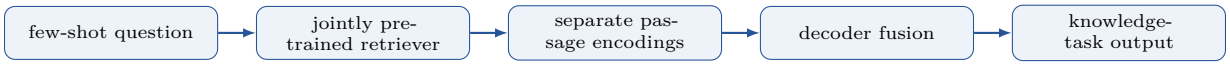
Motivation and mechanism. Atlas asks whether retrieval-augmented pretraining can make knowledge-intensive tasks data efficient, rather than treating retrieval only as an inference add-on [Izacard et al., 2022]. A dual-encoder retriever selects passages and a Fusion-in-Decoder reader encodes each question–passage pair separately before the decoder attends across all encoded passages. Retriever supervision can come from reader attention or leave-one-out evidence estimates. Abstractly,

$$z_{1:k} = \mathrm{TopK}_{z \in \mathcal{D}}\, s_\eta(q, z), \qquad p(y \mid q, z_{1:k}) = \mathrm{FiD}_\theta(q, z_{1:k}). \tag{37}$$

The external index can be updated without storing all knowledge in model parameters, while joint pretraining teaches the retriever and reader a shared task interface.

Listing 22: Atlas-style retrieve, encode separately, and fuse in the decoder

```
def atlas_forward(question, retriever, reader, corpus, k):
    passages = corpus.fetch(retriever.topk(question, k))
    encoded = [reader.encode(question, passage) for passage in passages]
    return reader.decode(torch.cat(encoded, dim=1))
```



Evidence and limitations. Atlas evaluates few-shot and full-data learning on Natural Questions, KILT tasks, MMLU, and related knowledge-intensive benchmarks. The paper also varies index content and retrieval-training objectives. Its contribution is strongest at the training interface: retrieval provides updateable evidence and sample-efficient task adaptation. It still pays to retrieve and encode a passage set before generation, and reader-derived retriever targets can reinforce the reader's own shortcuts.

PRA relation and experiment. Atlas supplies a stronger pretrained retrieval baseline than prompt-only RAG. A direct comparison should initialize both systems from the same corpus and language backbone, then vary the number of task examples. Atlas materializes passage encodings; PRA materializes selected per-layer chunk K/V. Match retrieved tokens, pretraining data, and online encoding cost. Oracle passages test transport, while valid-versus-shuffled evidence and index-refresh tests measure content and updateability.

6.5 Self-RAG

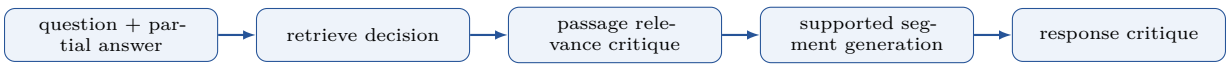
Motivation and mechanism. Fixed-k RAG retrieves even when parametric knowledge is sufficient and may accept irrelevant evidence without an explicit check. Self-RAG trains the generator to emit reflection tokens that control retrieval and assess passage relevance, support, and response quality [Asai et al., 2024]. A simplified policy alternates

$$a_t \in \{\textsc{retrieve}, \textsc{continue}\}, \qquad r_t = \mathrm{critique}(q, z_t, y_{<t}), \tag{38}$$

then uses critique scores to filter or rank candidate continuations. Retrieval triggering and evidence criticism therefore become learned generation decisions rather than fixed pipeline settings.

Listing 23: Simplified adaptive retrieval and reflection loop

```
while not finished:
    action = model.next_control_token(question, draft)
    if action == RETRIEVE:
        passages = retriever.search(question + draft)
        passages = rank_by_reflection(model, question, draft, passages)
    draft += model.generate_segment(question, draft, passages)
return draft
```



Evidence and limitations. Self-RAG evaluates open-domain QA, reasoning, fact verification, long-form factuality, and citation behavior with 7B- and 13B-scale models. Reflection tokens make retrieval frequency controllable, but their labels depend on a teacher and critic pipeline. A plausible self-critique can share the generator's errors, and discrete retrieve/generate cycles still incur text and serial decoding cost.

PRA relation and experiment. Self-RAG is the closest retrieval analogue to PRA's proposed progressive trigger. Compare a reflection-token trigger, a latent PRA gate, and a shared oracle trigger while holding the retriever and evidence fixed. Measure trigger precision/recall, unnecessary accesses, evidence support, serial latency, and content counterfactuals. A hybrid can expose PRA route confidence and provenance to a Self-RAG critic, but the critic's approval must not replace removal and substitution tests.

6.6 RETRO

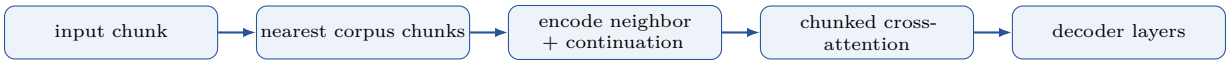
Motivation and mechanism. RETRO retrieves neighbors for fixed-size chunks from a massive token database, encodes each neighbor and its continuation, and interleaves chunked cross-attention in the decoder [Borgeaud et al., 2022]. For input chunk x_c and neighbors N_c,

$$N_c = \mathrm{kNN}(E(x_c), \mathcal{D}), \qquad H'_c = H_c + \mathrm{CrossAttn}(H_c, \mathrm{Enc}(N_c)). \tag{39}$$

Retrieval happens at chunk boundaries, bringing non-parametric data closer to layer computation than prompt-only RAG.

Listing 24: RETRO-style chunk retrieval and cross-attention

```
for chunk in split(tokens, chunk_len):
    ids = index.search(chunk_encoder(chunk), k=neighbors)
    neighbor_states = encoder(corpus.with_continuations(ids))
    h = decoder_chunk(chunk, retrieved=neighbor_states) # selected layers cross-attend
```



Evidence, ablations, and limitations. RETRO trains models with a database on the order of trillions of tokens and compares perplexity and downstream behavior against larger parametric Transformers. Retrieval neighbors, database scale, cross-attention placement, and corpus overlap are central ablations. Results support non-parametric scaling, but the system inherits corpus/index cost, possible benchmark leakage, chunk-boundary mismatch, and retrieval latency; source identity is incidental rather than an explicit user-facing contract.

PRA relation and experiment. RETRO is one of PRA's closest architectural ancestors: both transport external representations through cross-attention. PRA adds explicit handles, object boundaries, layer-specific reusable cache identity, progressive admission, and causal reference controls. Under one corpus and backbone, compare no handle, noisy handle, exact handle, and global semantic retrieval; keep returned tokens and PRA layers fixed. Measure selection recall, quality, per-stage latency, cache reuse, source corruption, and multi-object composition. Oracle selection isolates whether PRA's transport works before router claims.

7 Category V: Attention-Level Retrieval and KV Runtime Management

This category is easy to conflate because every method manipulates latent states. The systems nevertheless perform different operations. Unlimiformer and Landmark Attention select semantic states or blocks. PagedAttention virtualizes physical allocation. StreamingLLM, H_2O, Scissorhands, SnapKV, FastGen, PyramidKV, and Quest reduce which already-produced states are loaded or retained. MiniCache compresses across layers, while CacheBlend repairs reusable chunk caches that were encoded outside their new prompt position. Selection, allocation, retention, representation compression, and reuse should therefore be measured separately.

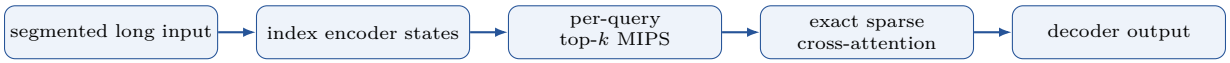
7.1 Unlimiformer

Motivation and mechanism. Unlimiformer lets a pretrained encoder-decoder process arbitrarily segmented input, indexes all encoder hidden states, and replaces dense decoder cross-attention with exact attention over MIPS-selected states [Bertsch et al., 2023]. Because the cross-attention score is an inner product, each decoder query can retrieve its largest candidate logits without first forming the full query–key matrix:

$$\mathcal{I}(q) = \mathrm{TopK}_i(q^\top k_i), \qquad o = \mathrm{softmax}(qK_{\mathcal{I}}^\top)V_{\mathcal{I}}. \tag{40}$$

Listing 25: Token-state MIPS followed by exact attention

```
def unlimiformer_cross_attention(q, index, value_store, k):
    scores, ids = index.search(q, k)
    values = value_store[ids]
    return torch.einsum('...k,...kd->...d', scores.softmax(-1), values)
```



Evidence and limitations. The paper evaluates long-input encoder-decoder tasks, especially summarization, against truncated, segmented, and long-context baselines using ROUGE and memory/length measures. It also studies training-free wrapping and fine-tuning. Selection occurs at token granularity and can miss jointly useful low-ranked tokens; the index is tied to the current encoded input; ANN setup and query cost remain; and source provenance must be reconstructed from token offsets.

PRA relation and comparison. Unlimiformer and PRA meet at sparse latent attention. Unlimiformer indexes all token states of the current long input, while PRA first selects named objects and may persist their per-layer KVs. Compare full cross-attention, Unlimiformer token MIPS, PRA object routing plus dense within-object attention, and a hierarchical object-to-token cascade. Match returned KVs, encoder FLOPs, and peak bytes; sweep object size, distractors, repeated queries, and object-level corruption. The cascade tests whether explicit identity improves efficiency without sacrificing token precision.

7.2 Landmark Attention

Motivation and mechanism. Ordinary sparse attention requires a fixed graph or an external retriever. Landmark Attention instead trains the model's own attention to use one special landmark token as the representative key for each token block [Mohtashami and Jaggi, 2023]. At inference, a query first scores landmarks, retrieves the top blocks, and then attends to token K/V inside those blocks. If λ_b is the landmark for block b,

$$\mathcal{B}(q) = \mathrm{TopK}_b(q^\top k_{\lambda_b}), \qquad o = \mathrm{Attn}(q, K_{\mathcal{B}(q)}, V_{\mathcal{B}(q)}). \tag{41}$$

The method preserves random access to distant blocks while avoiding attention over every token. Its optional context-miss token also suggests a learned decision not to retrieve.

Listing 26: Landmark routing followed by block attention

```
def landmark_attention(query, landmark_keys, block_keys, block_values, k):
    block_ids = (query @ landmark_keys.T).topk(k).indices
    keys = torch.cat([block_keys[i] for i in block_ids], dim=-2)
    values = torch.cat([block_values[i] for i in block_ids], dim=-2)
    return scaled_dot_product_attention(query, keys, values)
```

Evidence and limitations. The paper evaluates PG19 language modeling and passkey retrieval, including a LLaMA-7B adaptation beyond its original context. It studies block retrieval granularity and a context-miss threshold. Landmark keys are learned in the same attention space as their consumers, but every block must be ingested and assigned a landmark. Blocks remain positional rather than stable versioned objects, and a wrong landmark decision hides every token inside the missed block.

PRA relation and experiment. Landmark Attention is the closest token-stream analogue to PRA's corrected chunk gists. Both rank a small projected-key representative and transport detailed token K/V after selection. The differences are identity, lifecycle, and scope: landmarks index blocks of an ingested stream, whereas PRA gists belong to resolved URI objects that may be cached and invalidated. Compare both with identical chunk boundaries, top-k, and attention projections; then vary object reuse, reorder blocks, change versions, and measure cold versus warm construction. This experiment can reveal whether explicit object identity adds value beyond learned block routing.

7.3 PagedAttention

Motivation and mechanism. PagedAttention is a serving-memory design in vLLM, not a semantic attention rule. It partitions a sequence's KV cache into fixed-size physical blocks addressed through a logical block table, reducing fragmentation and enabling copy-on-write sharing [Kwon et al., 2023]. For logical token position t,

$$b = \lfloor t/B \rfloor, \quad o = t \bmod B, \quad (K_t, V_t) = \mathrm{blockTable}[b][o]. \tag{42}$$

Listing 27: Logical-to-physical KV lookup

```
def paged_lookup(logical_positions, block_table, blocks, block_size):
    logical_block = logical_positions // block_size
    offset = logical_positions % block_size
    physical_block = block_table[logical_block]
    return blocks[physical_block, offset]
```

Evidence, systems lineage, and failures. vLLM evaluates serving throughput, latency, memory utilization, and scheduling across model sizes and workloads against contemporary inference systems. Block size, batching, sharing, and scheduling are systems variables rather than accuracy ablations. Paging cannot decide which knowledge is useful; small blocks increase metadata, large blocks reintroduce internal fragmentation, and shared physical storage must preserve isolation and copy-on-write semantics.

PRA relation and experiment. The designs compose: PRA supplies semantic object identity and PagedAttention supplies allocation. Map an immutable (u, v, ℓ, m, p) object to reference-counted KV blocks and maintain per-sequence block tables/masks. Compare contiguous and paged PRA caches under cold, warm, and authorized cross-request reuse. Measure bytes allocated, fragmentation, blocks shared, avoided encoding, time to first token, throughput, eviction, and a confidentiality test where another sequence's private cache must leave the target logits invariant.

7.4 StreamingLLM

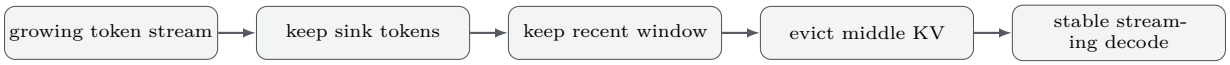
Motivation and mechanism. StreamingLLM observes that autoregressive models often assign disproportionate attention to initial "sink" tokens even when those tokens are not semantically important. Retaining a few initial KVs plus a rolling recent window allows stable streaming beyond the training context [Xiao et al., 2023]:

$$\mathcal{K}_t = \{0, \ldots, s-1\} \cup \{t-w+1, \ldots, t\}. \tag{43}$$

Middle-history KVs are evicted; no semantic retrieval or reconstruction is attempted.

Listing 28: Attention-sink plus rolling-window retention

```
def streaming_keep(length, sinks=4, window=1024):
    first = torch.arange(min(sinks, length))
    recent = torch.arange(max(sinks, length-window), length)
    return torch.unique(torch.cat([first, recent]))
```



Evidence and failure modes. The paper tests long streaming language modeling and generation with LLaMA-style models, comparing sliding windows and cache baselines through perplexity and long-run behavior. Sink tokens repair an attention-distribution pathology; they do not preserve arbitrary facts. Middle-history details are irrecoverable, sink behavior depends on model training, and the method is not a solution for non-chronological external knowledge.

PRA relation and experiment. StreamingLLM manages the local chronological cache; PRA retrieves named backing objects. Combine them by keeping sinks/recent discourse locally and dereferencing remote evidence. Under a fixed total KV budget, compare rolling window, sink+window, PRA, and the hybrid. Sweep query distance, explicit handles, and whether lost facts are still present in the backing store. Charge PRA faults and test shuffled content; test StreamingLLM with sink removal and exact middle-history probes.

7.5 H_2O: Heavy-Hitter Oracle

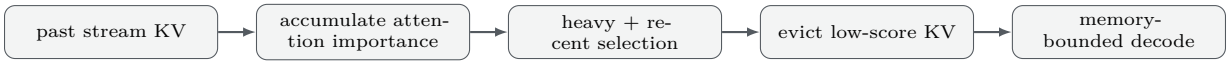
Motivation and mechanism. H_2O treats KV eviction as a dynamic submodular selection problem. It retains recent tokens and "heavy hitters" with large accumulated attention contribution [Zhang et al., 2023]. A simplified score is

$$I_i^{(t)} = I_i^{(t-1)} + \sum_h A_{h,t,i}, \qquad \mathcal{K}_t = \mathrm{Recent}_r \cup \mathrm{TopK}_h(I^{(t)}). \tag{44}$$

The policy reacts to observed decoder attention rather than using a static window.

Listing 29: Educational heavy-hitter update

```
def h2o_keep(importance, attn_to_past, recent_ids, heavy_budget):
    importance[:attn_to_past.numel()] += attn_to_past.detach()
    heavy = importance.topk(heavy_budget).indices
    return torch.unique(torch.cat([heavy, recent_ids])), importance
```



Evidence and limitations. H_2O evaluates OPT, LLaMA, and GPT-NeoX across language and generation tasks, measuring quality preservation, cache size, throughput, and latency against inference systems and eviction baselines. Its reported systems gains are conditional on model, hardware, batch, cache fraction, and implementation. Accumulated attention is a lagging proxy, early errors are irreversible, low-attention setup facts can matter later, and raw attention mass is not causal importance.

PRA relation and experiment. H_2O decides which states from one active stream remain; PRA decides which external objects enter. Use H_2O within each selected PRA object or across the resident object working set, but preserve URI/version metadata for every KV. Compare full object KVs, H_2O-compressed objects, object-level eviction, and both tiers. Sweep delayed-use facts, distractors, object count, and warm-cache reuse. Removal/substitution tests should verify that retained heavy hitters, rather than generic branch activation, cause correct answers.

7.6 Scissorhands

Motivation and mechanism. Scissorhands builds on the "persistence of importance" hypothesis: tokens important in earlier attention tend to remain important later. It maintains importance statistics and discards KVs predicted to stay unimportant [Liu et al., 2023]. At an eviction step,

$$\mathcal{K} = \text{Recent}_r \cup \text{TopK}_{B-r}(\text{score}_{\text{history}}(i)). \tag{45}$$

Policies can be applied per head/layer and periodically rather than at every token.

Listing 30: Periodic history-score pruning

```
def scissorhands(kv, score, recent, budget):
    protected = torch.arange(max(0, len(score)-recent), len(score))
    old = torch.arange(0, max(0, len(score)-recent))
    keep_old = old[score[old].topk(max(0, budget-len(protected))).indices]
    keep = torch.cat([keep_old, protected])
    return kv.index_select(-2, keep), score.index_select(0, keep)
```

Evidence and failure modes. The work evaluates cache reduction across language models/tasks using perplexity or task quality plus memory and throughput. Comparisons with window and heuristic eviction establish the value of history-aware scores. Importance persistence can fail under topic shifts or late references; dropped tokens cannot be reconstructed; per-head policies complicate kernels; and attention history remains a correlational diagnostic.

PRA relation and experiment. Scissorhands compresses already materialized stream history, whereas PRA can reload a stable object after eviction. Compare irreversible token eviction, eviction plus raw backing-text reload, and eviction plus cached PRA object KVs. Match total bytes and include late queries after long distractor gaps. Report reload rate, quality, end-to-end latency, and counterfactual content effects. Object boundaries may provide a cleaner eviction unit than anonymous tokens, but that is a testable hypothesis.

7.7 SnapKV

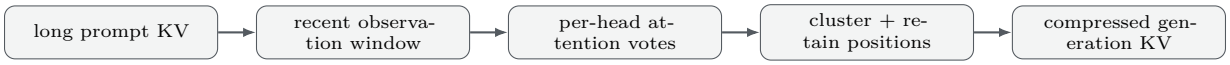
Motivation and mechanism. SnapKV compresses a long prompt's KV cache before generation by using attention from a recent observation window to identify prompt positions likely to matter during decoding, then pooling nearby selections [Li et al., 2024]. A schematic per-head rule is

$$s_i = \text{pool}\left(\sum_{q \in W_{\text{obs}}} A_{q,i}\right), \qquad \mathcal{K} = \text{TopK}_B(s). \tag{46}$$

Unlike online eviction, it takes a "snapshot" of prompt relevance before the answer grows.

Listing 31: Observation-window prompt selection

```
def snapkv_scores(attn, observation, pool=7):
    # attn: [heads, query, key]; exclude generated/recent keys as appropriate
    score = attn[:, -observation:, :].sum(dim=1).unsqueeze(1)
    return F.max_pool1d(score, kernel_size=pool, stride=1, padding=pool//2).squeeze(1)
```



Evidence and limitations. SnapKV evaluates long-context QA/retrieval and generation across instruction-tuned LLMs, measuring task quality, prompt-cache reduction, and speed against eviction/compression baselines. Its main assumption is that the observation window predicts future importance. Later reasoning can change the relevant region; prompt positions lose object/version identity; selection errors are irreversible unless the source remains elsewhere; and head-wise layouts complicate reuse.

PRA relation and experiment. SnapKV is query-aware token compression after eager prompt ingestion; PRA is object admission before or during latent use. Compare full prompt, SnapKV, PRA, and PRA with SnapKV-compressed object KVs. Match resident bytes and include questions whose relevance is evident initially versus revealed after intermediate reasoning. Measure prefill, selection, cold encoding, decode latency, exact evidence retention, and valid/shuffled object effects. A multi-fidelity PRA cache could store a SnapKV snapshot and escalate to full object KVs on low confidence.

7.8 FastGen and PyramidKV: Head- and Layer-Adaptive Budgets

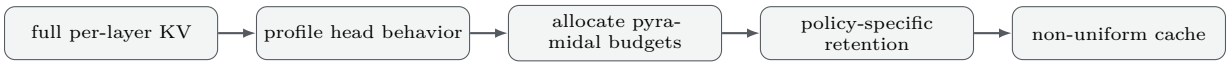
Motivation and mechanism. A uniform cache budget assumes that every attention head and layer uses history in the same way. FastGen profiles each head and assigns a retention policy that matches its observed structure: local heads keep a window, special-token heads keep designated positions, and broadly distributed heads retain more of the cache [Ge et al., 2023]. PyramidKV operates along a different axis. It reports that attention is broad in lower layers and more concentrated in higher layers, then uses a decreasing layer budget [Cai et al., 2024]. A generic allocation is

$$B_{\ell,h} = B_{\text{total}} \frac{w_{\ell,h}}{\sum_{j,g} w_{j,g}}, \tag{47}$$

where the weight $w_{\ell,h}$ comes from head profiling, layer depth, or both. The equation does not prescribe either paper's exact policy; it exposes their shared principle that cache capacity should follow attention structure rather than tensor shape.

Listing 32: Composing head policy with a layer-dependent cache budget

```
for layer, heads in enumerate(model.attention_heads):
    layer_budget = pyramid_budget(layer, model.num_layers, total_budget)
    for head in heads:
        policy = profiled_policy(head) # local, special-token, or broad
        head.cache = policy.retain(head.cache, budget=layer_budget[head.id])
```



Evidence and limitations. FastGen evaluates training-free adaptive compression across language-model tasks and reports memory savings with small quality changes under its tested settings. PyramidKV evaluates LongBench and needle retrieval, including aggressive retention fractions. Both depend on attention patterns transferring across inputs. A head classified as local can become globally important on another task, and a depth trend does not identify which individual evidence tokens must survive. Specialized per-head layouts can also reduce kernel regularity.

PRA relation and experiment. These methods can compress each selected PRA object's token K/V after object and chunk routing. Compare uniform and adaptive budgets under the same total bytes, with per-layer routing recall and content counterfactuals. The critical failure test delays a reference until a head or layer previously classified as low-value must retrieve it. A successful composition would reduce resident bytes without changing which URI and chunk caused the answer.

7.9 Quest: Query-Aware Page Selection

Motivation and mechanism. Retention policies decide what survives before the future query is known. Quest retains paged K/V metadata and decides which pages to load for the current query [Tang et al., 2024]. For each key

dimension, a page stores extrema $k_p^{\min}$ and $k_p^{\max}$. The query signs choose an upper bound on the page's possible inner product,

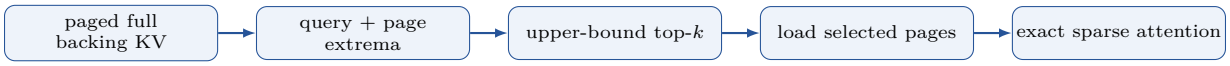
$$U(q,p) = \sum_j \max(q_j k_{p,j}^{\min}, q_j k_{p,j}^{\max}), \qquad \mathcal{P}(q) = \mathrm{TopK}_p\, U(q,p). \tag{48}$$

Only selected pages are transferred for exact attention. The method therefore reduces memory bandwidth without first discarding the full backing cache.

Listing 33: Quest-style query bound over KV pages

```
def quest_page_scores(query, page_key_min, page_key_max):
    lower = query * page_key_min
    upper = query * page_key_max
    return torch.maximum(lower, upper).sum(dim=-1)

pages = quest_page_scores(q, key_min, key_max).topk(page_budget).indices
output = exact_attention(q, load_keys(pages), load_values(pages))
```



Evidence and limitations. Quest reports self-attention speedups and end-to-end latency improvements on long-context tasks while preserving quality in the evaluated configurations. Its page bounds are inexpensive and query dependent, but loose bounds load irrelevant pages, tight budgets can omit diffuse evidence, and page layout determines selection granularity. The full cache still occupies a backing tier even when HBM traffic falls.

PRA relation and experiment. Quest performs physical page routing inside an already materialized context; PRA performs semantic object and chunk routing before transport. The two-stage composition is natural: PRA selects URI chunks and Quest selects pages inside their token K/V. Compare flat Quest over all candidate objects, hierarchical PRA-to-Quest, and PRA with dense selected chunks. Match page metadata, loaded bytes, and top-k tokens. Report both semantic recall and actual bandwidth, because a good router that loads loose pages may not improve the systems frontier.

7.10 MiniCache and CacheBlend: Depth Compression and Reusable Chunks

Motivation and mechanism. Token eviction compresses the sequence dimension, but KV redundancy also appears across depth and repeated prompts. MiniCache observes similarity between adjacent middle-to-late layers and merges corresponding states after separating vector direction from magnitude [Liu et al., 2024]. Distinct state pairs can remain unmerged. CacheBlend addresses a different reuse problem: a document chunk encoded alone does not contain the cross-chunk interactions it would receive in a new RAG prompt. It reuses offline chunk K/V but selectively recomputes important tokens to repair the blended context [Yao et al., 2024].

Listing 34: Two orthogonal cache transformations

```
# Depth compression: share similar adjacent-layer states.
merged = merge_direction_preserve_norm(kv[layer], kv[layer + 1], keep_distinct=True)

# Reuse repair: blend cached chunks, then recompute selected contextual tokens.
prompt_kv = concatenate(cached_chunk_kv)
repair_ids = select_recompute_tokens(prompt_kv, query)
prompt_kv[repair_ids] = model.prefill_tokens(prompt, repair_ids)
```

Evidence and limitations. MiniCache evaluates several LLaMA, Mistral, Phi, and Mixtral variants across language and long-context benchmarks, reporting compression, throughput, and quality. CacheBlend evaluates RAG serving with several open models and datasets, reporting time to first token and throughput against full recomputation and prefix reuse. Cross-layer similarity is model dependent, while chunk reuse is position and context dependent. In both cases, approximate equivalence must be tested on compositional questions, not inferred from cache similarity alone.

PRA relation and experiment. MiniCache can reduce the depth cost of a PRA object; CacheBlend is a direct baseline for PRA's reusable-object systems claim. Compare full per-request encoding, prefix-only reuse, CacheBlend repair, position-independent PRA cross-attention, and PRA with depth-compressed object K/V. Use reordered documents, multi-document composition, stale versions, and shared-cache workloads. Measure answer quality, repair fraction, encoding avoided, bytes transferred, time to first token, and isolation. PRA should not claim reusable K/V advantage unless it beats these stronger reuse baselines at matched quality.

7.11 KV management design map

The methods above can be compared by the unit they control and by whether omitted detail still exists in a backing tier. That distinction predicts recoverability: page selection can load another page on the next query, whereas destructive token eviction cannot recover a dropped state unless raw content is re-encoded.

Table 2: KV-management mechanisms control different logical and physical units.

Method	Controlled unit	Decision time	Backing detail	Training change	Principal risk
PagedAttention	physical block	allocation	full K/V remains	no	fragmentation or isolation bug
StreamingLLM	token positions	decode	evicted middle absent	no	unrecoverable old detail
H_2O	token positions	decode	evicted tokens absent	no	lagging attention score
Scissorhands	token/head K/V	periodic decode	evicted tokens absent	no	importance changes later
SnapKV	prompt token/head	before generation	source may remain external	no	observation window predicts poorly
FastGen	head policy	profile/prefill	policy dependent	no fine-tuning	profile fails to transfer
PyramidKV	layer budget	prefill	omitted tokens absent	no	depth heuristic misses evidence
Quest	K/V page	each query	full backing K/V remains	no	loose bounds or page miss
MiniCache	adjacent layers	prefill/decode	selected exceptions remain	no	cross-layer mismatch
CacheBlend	reusable text chunk	prefill	raw text and offline K/V remain	no	insufficient contextual repair
Landmark Attention	token block	each query	block K/V remains	yes	wrong landmark hides block
PRA	URI chunk, then token K/V	layer/step	raw/versioned object may remain	yes/runtime policy	wrong route, stale identity

The map exposes three promising compositions. PagedAttention can allocate any of the logical caches. Quest can reduce bandwidth inside a selected Landmark or PRA block. MiniCache or an adaptive budget can compress the resulting resident tensors. CacheBlend, by contrast, is a direct alternative when the goal is reuse of text chunks inside ordinary self-attention. End-to-end comparisons must include all metadata, repair, transfer, and re-encoding costs; nominal cache bytes alone do not identify the faster system.

8 Category VI: Structural and Hierarchical Indexing

Flat chunk retrieval treats document structure as noise. Hierarchical and graph approaches make sentences, sections, entities, communities, or claims explicit intermediate units. Their advantage is not merely fewer tokens: the index supplies a reasoning topology and a natural provenance coordinate system.

8.1 Hierarchical Attention Networks (HAN)

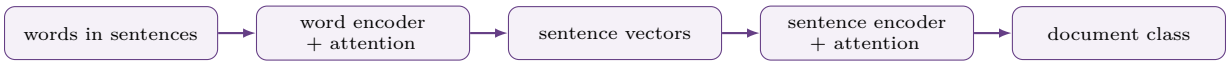
Motivation and mechanism. HAN mirrors document composition: a bidirectional word encoder and attention pool create sentence vectors, then a sentence encoder and attention pool create a document vector [Yang et al., 2016]. At either level,

$$u_i = \tanh(Wh_i + b), \quad \alpha_i = \frac{\exp(u_i^\top u_c)}{\sum_j \exp(u_j^\top u_c)}, \quad v = \sum_i \alpha_i h_i. \tag{49}$$

The context vector u_c learns which units matter for classification. Hierarchy reduces a variable document to semantically meaningful intermediate representations.

Listing 35: Reusable hierarchical attention pool

```
class AttnPool(nn.Module):
    def __init__(self, d):
        super().__init__(); self.proj = nn.Linear(d, d); self.context = nn.Parameter(torch.randn(d))
    def forward(self, h, mask=None):
        score = torch.tanh(self.proj(h)) @ self.context
        if mask is not None: score = score.masked_fill(-mask, -torch.inf)
        return torch.einsum('bt,btd->bd', score.softmax(-1), h)
```



Evidence and limitations. The original paper evaluates document classification on Yelp, IMDB, Yahoo Answers, and Amazon review datasets against bag-of-words, convolutional, and recurrent baselines using classification accuracy. Word- and sentence-level attention visualizations are useful diagnostics, not causal explanations. The hierarchy is fixed at two levels, pooling is lossy, bidirectional encoding is not a streaming solution, and the model is designed for one document-level prediction rather than open-ended retrieval.

PRA relation and experiment. HAN supplies the principle that intermediate document units deserve representations. PRA generalizes those units to named sections, tables, records, and code symbols and permits later expansion. Compare flat chunks, HAN-style sentence summaries, and URI hierarchy on classification plus evidence-required QA. Remove or substitute high-attention sentences and selected PRA nodes. A strong hybrid uses HAN pooling to construct cheap summaries while preserving children as resolvable objects.

8.2 RAPTOR

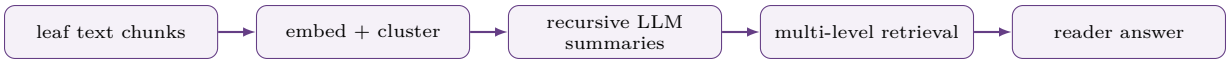
Motivation and mechanism. RAPTOR recursively embeds and clusters leaf chunks, asks a language model to summarize each cluster, and repeats until a tree of increasingly abstract nodes is formed [Sarthi et al., 2024]. With nodes V_0 at the leaves,

$$V_{\ell+1} = \{\text{summarize}(C) : C \in \text{cluster}(\{E(v) : v \in V_\ell\})\}. \tag{50}$$

Retrieval can search a collapsed set of all levels or traverse the tree, letting a query choose between detailed evidence and broad synthesis.

Listing 36: Bottom-up RAPTOR index construction

```
level = chunk(document)
tree = [level]
while len(level) > 1:
    groups = cluster(embed(level))
    level = [summarizer(group) for group in groups]
    tree.append(level)
return tree
```



Evidence, paper trail, and failures. RAPTOR reports QA results on QuALITY, NarrativeQA, and QASPER, comparing conventional retrieval and long-context/retrieval baselines with accuracy or task metrics. The paper highlights a 20-point absolute QuALITY gain for its GPT-4 configuration; that number is conditional on its model, tree, prompts, and evaluation. Descendants study dynamic maintenance and cheaper tree construction. Indexing is expensive, abstractive summaries can omit or invent facts, clustering is unstable, and an error at a high node can misroute an entire subtree.

PRA relation and experiment. RAPTOR chooses representations during indexing; PRA chooses when and at which layer a representation becomes resident. Assign immutable URIs to every tree node and let the router escalate from summary to children. Compare flat RAG, collapsed-tree retrieval, explicit tree traversal, and PRA transport using the same tree. Measure answer quality, retrieved source tokens, LLM index cost, expansions, cache reuse, summary corruption, and whether leaf evidence supports generated claims.

8.3 MemWalker

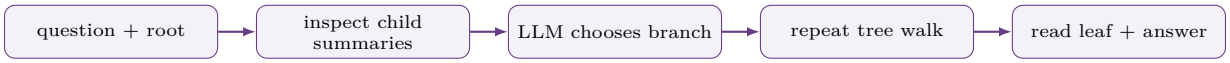
Motivation and mechanism. MemWalker recursively summarizes a long document into a memory tree and has an LLM navigate it by choosing children until sufficient evidence is found [Chen et al., 2023]. At node v_t, a policy reads the query and child summaries:

$$a_t \sim \pi_\theta(a \mid q, s(v_t), \{s(c) : c \in \text{child}(v_t)\}), \qquad v_{t+1} = \text{child}(v_t, a_t). \tag{51}$$

The traversal trace externalizes intermediate access decisions instead of retrieving all nodes in one ranking call.

Listing 37: Bounded tree walk

```
node = root
for depth in range(max_depth):
    action = navigator.choose(question, node.summary,
                              [c.summary for c in node.children])
    if action == "answer": break
    node = node.children[action]
return reader(question, node.source_text)
```



Evidence and failure modes. MemWalker evaluates long-context reasoning/QA against retrieval and long-context baselines, measuring answer quality and traversal behavior. It is closer to an agent than static RAG, but the action space is confined to a prebuilt tree. Serial LLM calls add latency, greedy early choices can make relevant siblings unreachable, summary errors compound, and navigation traces show decisions without proving causal use of the selected leaf.

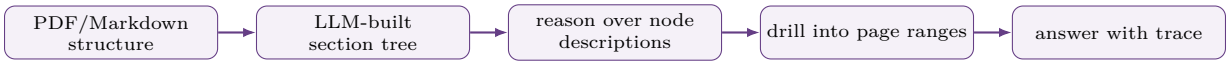
PRA relation and experiment. Both implement progressive disclosure. MemWalker serializes branch decisions as language-model actions; PRA proposes a latent router inside selected layers. Use the same tree and compare textual navigation, latent top-k child selection, oracle paths, and a hybrid that escalates ambiguous nodes to the agent. Match node-expansion budgets and report latency, depth, backtracking, path recall, valid/shuffled leaf effects, and calibration at the escalation boundary.

8.4 PageIndex

Motivation and mechanism. PageIndex builds a table-of-contents-like tree from natural document sections and uses an LLM to reason over titles, summaries, and page ranges rather than relying on vector similarity [Zhang et al., 2025]. A typical node is $(\text{id}, \text{title}, [p_s, p_e], \text{summary}, \text{children})$; retrieval is a reasoned tree search followed by page/section extraction.

Listing 38: Reasoning-first section traversal

```
frontier = [root]
while budget.remaining() and not evidence_sufficient(frontier, query):
    decision = llm_rank(query, describe(frontier))
    frontier = expand_selected(frontier, decision.node_ids)
pages = merge_ranges([n.page_range for n in frontier])
return answer(query, pdf.extract(pages))
```



Evidence status and reproducibility. PageIndex is an evolving open-source/product system, not a settled peer-reviewed architecture result. This survey pins its inspection target to repository commit `7592163e2a37` (mirror snapshot synchronized 15 June 2026) and records the repository's then-documented default `gpt-4o-2024-11-20`; access date is 3 August 2026. The repository advertises 98.7% FinanceBench accuracy, but this remains a vendor-reported compound-system result. Reproduction must freeze PDFs/questions, parser/OCR, tree-building and answering prompts, model versions, judge, exact-match policy, retries, and dollar/token cost. Natural structure can itself be missing or malformed, and every LLM tree call can introduce nondeterminism.

PRA relation and experiment. PageIndex supplies a strong logical namespace for PRA: node IDs/page ranges become versioned URIs, and selected sections can materialize as text or per-layer KV. Compare vector RAG, PageIndex traversal, PRA latent traversal, and agent escalation on the same tree. Stratify exact lookup, cross-reference, multi-section synthesis, and cross-document reasoning. Report source-page recall, answer/citation quality, tree calls, latency, cost, and invalidation after a PDF revision.

8.5 GraphRAG

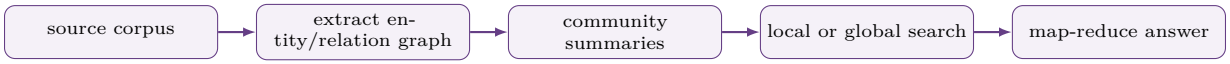
Motivation and mechanism. GraphRAG extracts entities, relations, and claims from a corpus, clusters the resulting graph into communities, and creates community reports [Edge et al., 2024]. Local search expands around query entities; global search maps a question over community reports and reduces partial answers:

$$\hat{y} = \text{reduce}\{\text{answer}(q, R_c) : c \in \text{select}(q, \mathcal{C})\}. \tag{52}$$

The graph exposes corpus-wide themes and multi-hop relationships that may not be similar to the surface query.

Listing 39: GraphRAG global map-reduce sketch

```
reports = select_community_reports(query, graph.communities, token_budget)
partials = [llm.map_answer(query, report) for report in reports]
return llm.reduce_answer(query, partials)
```



Evidence and failure modes. The GraphRAG preprint studies global sensemaking over roughly million-token corpora and reports LLM-judged comprehensiveness and diversity relative to conventional RAG. These are inference-time judgments over an expensive LLM-derived index, so model, prompts, judge, and cost are integral conditions. Extraction errors, entity resolution, hallucinated relations, index cost, stale derived summaries, and judge bias are prominent risks. Local factual QA and global thematic synthesis are distinct tasks.

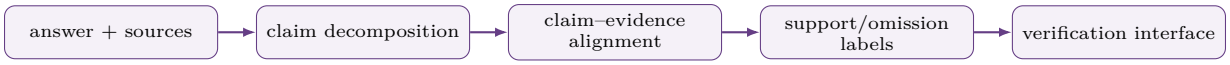
PRA relation and experiment. A PRA URI can address a node, neighborhood, graph query result, or community report. The hard problem becomes dependency-aware invalidation: changing one source may stale many derived KVs. Compare flat RAG, GraphRAG text transport, and graph-node PRA transport on local and global questions. Match source and answer models; report index cost separately from per-query cost, provenance to original spans, stale-update tests, and counterfactual removal of decisive nodes.

8.6 PaperTrail as provenance and HCI

Motivation and mechanism. PaperTrail is not a context architecture. It decomposes an answer and its sources into claims and evidence, then visualizes supported claims, unsupported assertions, and omitted source information [Martin-Boyle et al., 2026]. If C is answer claims and E source evidence units, the interface exposes a relation $R \subseteq C \times E$ plus unsupported $C \setminus \mathrm{dom}(R)$ and uncovered source claims.

Listing 40: Claim–evidence audit pipeline

```
answer_claims = decompose_claims(answer)
source_claims = decompose_claims(sources)
links = align_and_verify(answer_claims, source_claims)
render(answer_claims, source_claims, links,
       unsupported=set(answer_claims)-links.claims())
```



Evidence and limitations. The 2026 work evaluates PaperTrail in a within-subjects study with 26 researchers performing scholarly editing tasks. The interface lowered trust relative to a baseline but did not produce corresponding behavioral change: participants still relied on generated edits when verification was cognitively costly. This is an HCI result about provenance presentation, not evidence for longer context or retrieval quality. Claim decomposition and alignment can themselves be wrong, and more visible evidence can overload rather than assist a user.

PRA relation and experiment. PRA's URI/version and routing traces can populate a PaperTrail-like view, but attended memory is not causal support. Record selected objects, source spans, cache versions, generated claims, and outputs under removal/substitution. Compare citation-only, route-trace, and counterfactual claim-evidence interfaces. Measure verification accuracy, time, calibrated trust, unsupported-claim detection, omissions, and cognitive load. This positions PaperTrail correctly as provenance/HCI rather than a competitor in the context architecture table.

9 Category VII: Agents with Search and Tool Use

Agents move relevance decisions outside a single forward pass. They can discover unknown sources, reformulate a question after observing evidence, act on the world, and backtrack. Their generality costs serial latency, tokens, tool failures, and a larger audit surface. PRA should be compared as a restricted fast path for repeated read-only access, not as a replacement for open-ended action.

9.1 ReAct

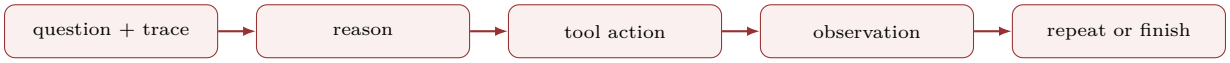
Motivation and mechanism. ReAct interleaves natural-language reasoning with actions and observations, allowing evidence acquisition to change the next reasoning step [Yao et al., 2023]. A trajectory is

$$\tau = (q, r_1, a_1, o_1, r_2, a_2, o_2, \ldots, y), \qquad (r_t, a_t) \sim \pi_\theta(\cdot \mid q, \tau_{<t}). \tag{53}$$

Reasoning maintains a task state; tools ground or alter that state; the policy decides when to stop. Search is only one environment—the paper also studies interactive tasks.

Listing 41: ReAct execution loop with an explicit budget

```
trace = []
for step in range(max_steps):
    thought, action = model.next(question, trace)
    if action.name == "finish": return action.answer, trace
    observation = tools[action.name](**action.arguments)
    trace.append((thought, action, observation))
raise BudgetExceeded(trace)
```



Evidence, descendants, and failures. ReAct evaluates HotpotQA and FEVER plus ALFWorld and WebShop, comparing standard prompting, chain-of-thought, act-only, and ReAct using exact match, factual accuracy, or environment success. Prompt examples, base model, tool interface, and action budget materially condition results. Descendants add planning, reflection, memory, and richer tools. Common failures are reasoning loops, malformed calls, irrelevant searches, premature stopping, prompt injection in observations, and high latency.

PRA relation and experiment. Both progressively acquire context. ReAct remains necessary when sources are unknown, actions have side effects, or explicit planning and backtracking are essential. Compare ReAct search, PRA over a known reference graph, and a tiered policy that tries local context, cached URI memory, structural navigation, then search. Match evidence/action budgets and report answer quality, calls, tokens, latency, provenance, cache reuse, and escalation calibration. Cache successful read-only discoveries as versioned references, never silently replay side-effecting actions.

9.2 Self-Ask with Search

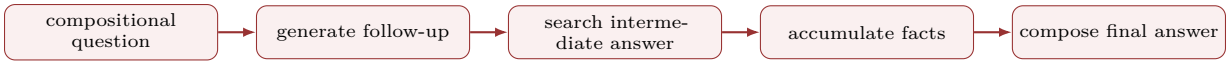
Motivation and mechanism. Self-Ask addresses compositionality by explicitly generating follow-up questions. An external search component answers each subquestion, and the model combines intermediate answers into the final response [Press et al., 2022]:

$$q_t = \text{decompose}(q, a_{<t}), \quad e_t = \text{search}(q_t), \quad y = \text{compose}(q, \{q_t, e_t\}). \tag{54}$$

The textual protocol makes the retrieval chain easy to inspect and separates decomposition errors from search errors.

Listing 42: Self-Ask protocol

```
facts = []
while model.needs_followup(question, facts):
    subquestion = model.followup(question, facts)
    evidence = search(subquestion)
    facts.append((subquestion, evidence))
return model.final_answer(question, facts)
```



Evidence and limitations. The work studies multi-hop/compositional question answering and the "compositionality gap", comparing direct answer, chain-of-thought, and search-assisted decomposition with exact match. Decomposition improves observability, but one wrong subquestion can derail all later hops; intermediate search snippets may be noisy; serial calls cost latency; and a plausible intermediate answer can conceal missing evidence.

PRA relation and experiment. PRA may represent a follow-up target as a generated reference only when a resolver can bind it deterministically; free-form web search is not a URI lookup. Compare text subquestions, latent reference queries, and a hybrid that uses PRA for known namespaces and search otherwise. Use the same search backend and hop budget; measure decomposition accuracy, evidence recall, calls, latency, valid/shuffled reference effects, and failures where the correct source is absent from the advertised graph.

9.3 IRCoT

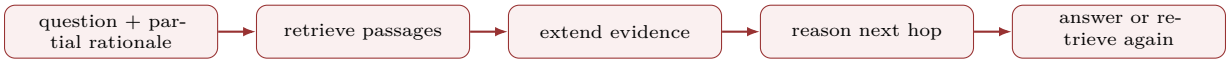
Motivation and mechanism. IRCoT interleaves retrieval with chain-of-thought so partial reasoning becomes a better query for the next evidence hop [Trivedi et al., 2023]. Rather than retrieve once from the original question,

$$z_t = R(q, r_{<t}, e_{<t}), \qquad r_t = G(q, r_{<t}, e_{<t}, z_t), \tag{55}$$

and the cycle continues until an answer or budget limit. Retrieval and reasoning therefore co-evolve.

Listing 43: IRCoT-style retrieval/reasoning alternation

```
state, evidence = question, []
for hop in range(max_hops):
    passages = retriever.search(state, k)
    evidence.extend(passages)
    state, answer = reasoner.step(question, state, passages)
    if answer is not None: return answer, evidence
```



Evidence, paper trail, and failures. IRCoT evaluates multi-step QA including HotpotQA, 2WikiMultiHopQA, MuSiQue, and related datasets, comparing retrieve-then-read, iterative retrieval, and chain-of-thought configurations with exact match/F1 and retrieval metrics. It motivates later iterative and agentic RAG. Reasoning text can introduce false entities that poison later queries, cost grows by hop, stopping is difficult, and answer gains can reflect a stronger prompt rather than a better retriever.

PRA relation and experiment. PRA layers or decoding states could emit successive latent queries, approximating IRCoT without serializing each query. Compare textual query rewriting and latent queries against the same frozen retriever, plus oracle evidence order. Match retrieval calls/candidates and report answer quality, hop recall, latency, and counterfactual evidence effects. The hybrid should expose an auditable object path even if queries are latent; otherwise reduced tokens come at the cost of provenance.

9.4 Toolformer

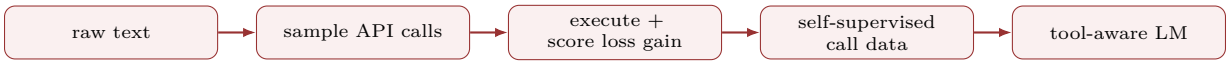
Motivation and mechanism. Toolformer teaches a language model when and how to call APIs by sampling candidate calls and retaining those that reduce language-model loss [Schick et al., 2023]. For call c inserted at position i, a simplified filter is

$$\Delta(c) = L(x_{i:} \mid x_{<i}) - L(x_{i:} \mid x_{<i}, c, \text{result}(c)); \quad c \text{ kept if } \Delta(c) > \tau. \tag{56}$$

Fine-tuning on retained demonstrations yields discrete call triggering and result integration without dense human call labels.

Listing 44: Loss-improvement filtering for pseudo-labeled calls

```
for position in candidate_positions(text):
    for call in sample_calls(model, text, position):
        result = execute_sandboxed(call)
        gain = loss_without(call) - loss_with(call, result)
        if gain > threshold: training_examples.append(insert(text, call, result))
```



Evidence and limitations. Toolformer evaluates a 6.7B GPT-J model with calculator, QA, search, translation, and calendar tools on zero/few-shot downstream tasks and language modeling, comparing base and tool-aware models. Tool choice and API formatting ablations matter. Pseudo-labels inherit the base model's blind spots, unsafe or costly calls require sandboxing, tool errors enter the context, and lower token loss does not guarantee factual or task utility.

PRA relation and experiment. Toolformer's call-retention principle is directly useful for PRA router training: sample candidate object accesses, keep those whose controlled insertion lowers loss, then train a gate with corrupted negatives. Compare heuristic cosine, supervised relevance, and loss-filtered routers under equal trainable parameters. Evaluate selection recall, calibration, task loss, valid/shuffled/irrelevant effects, and access cost. PRA can internalize deterministic reads; arbitrary or side-effecting APIs should remain explicit tool calls.

9.5 WebGPT and browser agents

Motivation and mechanism. WebGPT trains a model to issue search queries, open results, navigate pages, collect quotations, and answer with citations, using demonstrations, behavior cloning, reward modeling, and human feedback [Nakano et al., 2021]. The browser state makes information acquisition a sequential decision process:

$$a_t \sim \pi_\theta(a \mid s_t), \quad s_{t+1} = \text{browser}(s_t, a_t), \quad y = G(q, \text{quotes}(\tau)). \tag{57}$$

Listing 45: Citation-preserving browser loop

```
state, evidence = browser.start(question), []
for _ in range(action_budget):
    action = policy(state, evidence)
    if action.kind == "quote": evidence.append(browser.quote(action.span))
    elif action.kind == "answer": return compose_with_citations(question, evidence)
    else: state = browser.execute(action)
```

Evidence, descendants, and failures. WebGPT evaluates long-form question answering using human preference, factuality, and citation-related judgments against retrieval and human-written baselines. It established a lineage toward browser-assisted research agents. Results are bound to the web snapshot, search engine, model, demonstrations, reward model, and evaluator. Failures include source-quality errors, selective quotation, citation drift, prompt injection, browsing loops, changing pages, and high serial latency.

PRA relation and experiment. PRA cannot discover an unknown open-web source. It can cache already discovered pages, versions, sections, and evidence for later questions. Compare fresh browser search, cached text RAG, cached PRA KVs, and a policy that searches on cache miss or staleness. Measure answer/citation correctness, source diversity, actions, latency, cache hit and avoided encoding, update detection, and security under malicious page instructions. Provenance must retain the original page/version even when KVs are reused.

9.6 Recursive Language Models and context as environment

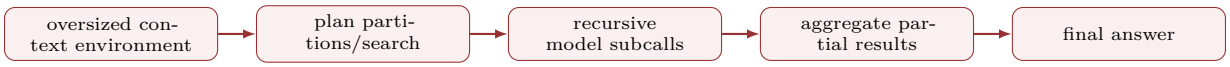
Motivation and mechanism. Recursive language-model approaches treat an oversized context as an external environment. A model writes code or issues subcalls that inspect, partition, summarize, and recursively solve pieces instead of consuming the entire source in one forward pass. For context D and budget B,

$$y = F(q, D, B) = \text{aggregate}\{F(q_j, D_j, B_j)\}_{j=1}^{m}, \qquad \sum_j B_j \le B. \tag{58}$$

The model controls the decomposition and can adapt it to each question.

Listing 46: Budgeted recursive context program

```
def solve(query, view, budget):
    if view.fits() or budget.depth == 0:
        return model(query, view.read(budget.tokens))
    plan = model.plan_partitions(query, view.metadata(), budget)
    partials = [solve(p.subquery, view.slice(p.range), budget.child(p)) for p in plan]
    return model.aggregate(query, partials)
```

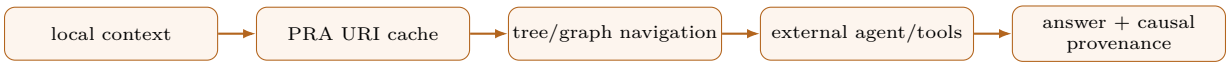


Evidence and failure modes. The family is evaluated on inputs beyond a model's window, code/document analysis, aggregation, and long-context reasoning, with answer quality, subcall count, tokens, and cost. Baselines should include truncation, long context, RAG, and non-recursive map-reduce. Generality is high, but recursive calls are expensive, partition boundaries can destroy dependencies, summaries compound errors, arbitrary generated code needs sandboxing, and budgets/stopping rules can dominate outcomes.

PRA relation and experiment. PRA can be viewed as a differentiable cached fast path for recurring read operations discovered by an agent or recursive program. Use identical source partitions and compare RLM subcalls, tree navigation, PRA references, and a tiered system that escalates on uncertainty. Match total model FLOPs/tokens and wall-clock budget; report depth, calls, source coverage, provenance, cache reuse, and tasks whose relevant partition is unknown initially. The agent should win on open-ended discovery; PRA must justify itself through repeated low-latency reuse and causal content selection.

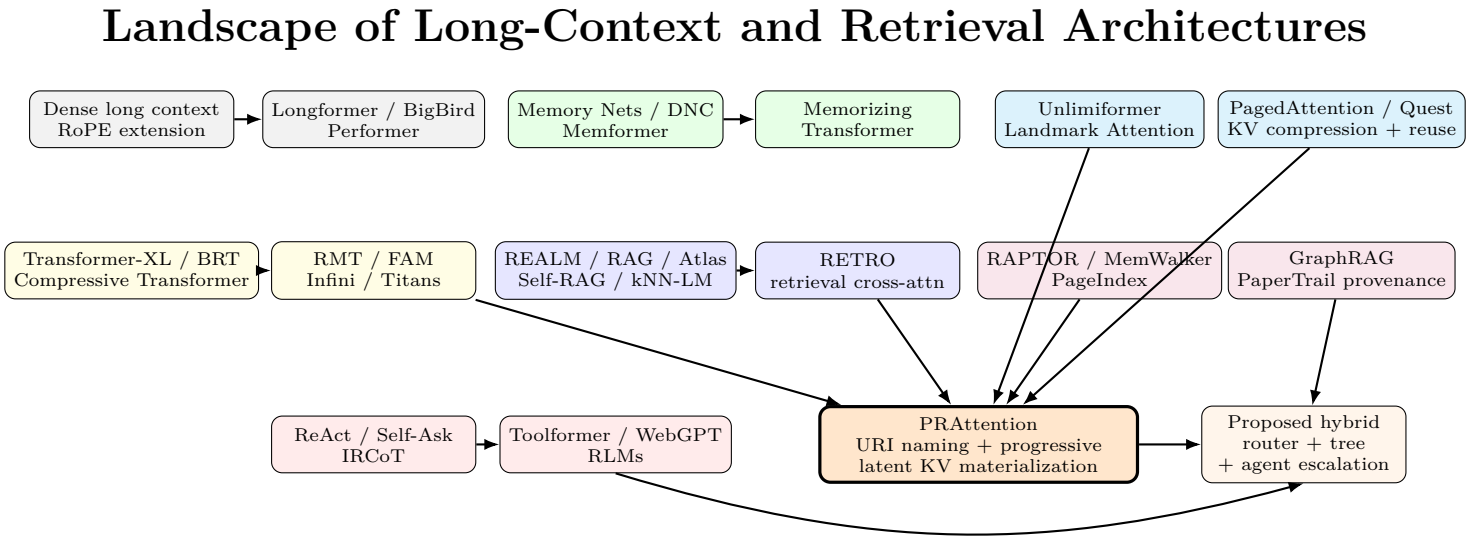
9.7 A tiered agent–PRA design

The categories are best composed as a monotone escalation policy. Local attention handles dense immediate context; a warm URI cache handles repeated named reads; a tree/graph index handles bounded structural search; and a general agent handles discovery or action. Each tier returns both evidence and calibrated confidence. Expensive discoveries may be cached only if the source is immutable or versioned, authorization permits reuse, and the action is read-only. The evaluation must include wrong-confidence costs: unnecessary escalation, silent cache hits on stale data, and premature stopping before an agent search.



10 One-Page Architecture Landscape

The preceding method descriptions can obscure a simple pattern: architectures differ less by whether they "use memory" than by when they pay to expose information and what identity survives that exposure. The landscape below places the seven families on those common axes. Arrows indicate useful conceptual transitions, not a chronological claim or a relation of strict inheritance.



Main axes: eager ↔ demand-driven access; flat tokens ↔ named structure; text ↔ latent KV; static one-shot selection ↔ progressive reasoning; transient context ↔ reusable cache; opaque use ↔ causal provenance.

11 Cross-Architecture Comparison

The comparison in this section treats each family as an access contract. The *store* column identifies what remains outside the current computation, while *selection time* identifies when the system narrows that store. The *form* column then records what crosses the boundary into the model. These distinctions matter because two systems can retrieve the same source yet incur different costs and preserve different evidence when one inserts text and the other retrieves K/V.

Family	**Store**	**Selection time**	**Form**	**Main strength**	**Characteristic failure**
Dense long context	prompt tokens	before forward pass	token KVs	general interactions	quadratic cost, distraction
Sparse attention	prompt tokens	architecture or task mask	sparse token KVs	long visible sequence	missing sparse paths
Recurrence	prior segments	chronological	hidden/KV state	streaming continuity	lossy or anonymous history
Test-time neural memory	observed stream	per token/block	updated latent parameters	adaptive long-term state	contamination, reset, correction
Differentiable memory	learned slots/episodes	per layer/step	latent vectors	trainable read/write	optimization and identity
Fixed RAG	document corpus	before or per-token generation	retrieved text encodings	fresh, updateable knowledge	chunking and retrieval miss
Adaptive RAG	document corpus	generated control decisions	text plus reflection state	learned retrieval timing	serial cost, self-critique error
RETRO	token chunks	chunk boundary	cross-attention encodings	scale via nonparametric data	index and corpus dependence
KV/landmark retrieval	hidden/KV index	attention time	selected states or pages	query-aware latent access	selection miss, weak identity
KV retention/reuse	active or stored KV	runtime stages	pruned, merged, or repaired KV	lower bytes and avoided prefill	irreversible loss, context mismatch
Tree/graph index	structural nodes	iterative or one-shot	summaries, spans, or subgraphs	multigranular reasoning	index construction errors
Search agent	open tools/web	iterative decisions	observations or text	maximum flexibility	latency, cost, tool errors
PRA	URI objects and K/V caches	layer/step, demand-driven	selected chunk token K/V	identity, reuse, progressive access	router learning, stale caches, runtime complexity

No row dominates every column. Dense context provides the broadest interaction graph once tokens are local, whereas retrieval and navigation avoid loading much of the store. Agents can discover sources outside a fixed index but pay for iterative model and tool calls. Persistent-state systems avoid external lookup but compress history into anonymous latents. KV policies can reduce bandwidth or prefill without solving source discovery. PRA is most directly compared with Landmark Attention, latent retrieval, CacheBlend-style reuse, and hierarchical indexing: its URI identity and chunk routing are useful only if they improve selection, reuse, or auditability enough to repay cache construction and control-plane complexity.

12 Adapting Original Experiments to PRA

12.1 A benchmark matrix

The repository example exposes several separable abilities: preserving ordinary language modeling, locating a named module, selecting the relevant method within it, and discovering an unadvertised source when the named objects are insufficient. A benchmark that tests only one ability cannot distinguish the access contracts above. A credible program therefore combines:

1. **Language preservation**: WikiText-2/103 or C4 slices before and after memory adaptation.
2. **Exact long-range retrieval**: passkeys, variable needles, UUID/value lookup, and repeated distractors.

3. **Natural long-document understanding**: NarrativeQA, Qasper, QuALITY, GovReport, arXiv/PubMed summarization, and LongBench-style tasks.
4. **Open-domain retrieval**: Natural Questions, TriviaQA, HotpotQA, 2WikiMultiHopQA, MuSiQue.
5. **Structured documents**: FinanceBench, legal filings, technical manuals, code repositories, and table/figure references.
6. **Agentic discovery**: web or corpus search tasks where the correct source is initially unknown.

The sequence is intentional. Language preservation is a compatibility gate. Synthetic retrieval isolates routing under known ground truth. Natural documents test whether the mechanism survives realistic discourse. Open-domain and agentic tasks should be added only when source discovery, rather than access to an advertised object, is itself under study.

12.2 Common conditions

For each task, the baseline set should include local-only SelfAttention, longer-context SelfAttention, sparse attention, recurrent or FAM state, one-shot RAG, Atlas-style trained retrieval, Self-RAG-style adaptive retrieval, Landmark or latent token retrieval, PRA with explicit handles, structural-tree retrieval, and agentic search. Systems experiments should additionally compare full K/V, query-aware page selection, adaptive retention, CacheBlend-style reuse, and PRA object caches. PRA with learned source discovery is a separate system variant because it removes the advertised-object assumption.

Comparisons should hold the corpus, tokenizer, generation model, evaluation prompts, and random data splits fixed where meaningful. They should report both equal visible evidence budgets and equal end-to-end compute budgets. Matching only parameter count is insufficient when one method performs retrieval, encoding, or extra model calls outside the measured forward pass.

12.3 PRA-specific causal matrix

Task improvement does not by itself show that a reference mechanism selected or used the right evidence. The interventions below alter one part of the access chain at a time. For example, shuffled content preserves branch activation and approximate tensor shape while breaking the association between a handle and its evidence.

Condition	What it tests	Expected interpretation
Valid	complete system	task utility
Disabled branch	any memory contribution	Δ_{use}
Shuffled content	semantic dependence	$\Delta_{content}$
Irrelevant matched length	distractor resistance	router specificity
Random URI, same content	identity dependence	naming shortcut
Same URI, wrong version	cache invalidation	staleness safety
Oracle object	selection upper bound	router headroom
All evidence local	access overhead	efficiency frontier
Agent oracle search	discovery upper bound	unresolved-source headroom

The conditions are most informative when crossed rather than reported independently. If valid and shuffled references perform alike, the memory branch may be acting as an adapter rather than carrying content. If an oracle object helps but learned routing does not, representation and fusion remain plausible while selection is the bottleneck. If even the oracle fails, training or materialization should be repaired before scaling the router.

12.4 Metrics beyond task accuracy

Report task quality, retrieval recall, ranking, reference causality, preservation, and systems cost together. Selection metrics should use candidate-count-aware chance rates and distinguish reference recall from within-reference chunk, landmark-block, and KV-page recall. Triggering systems should report access precision and unnecessary retrievals. Causal metrics should report loss or answer changes under removal and substitution, not attention weight alone. Systems measurements must separate resolve time, encode time, cache lookup, routing, attention, generation, and synchronization in cold, warm, and shared-cache regimes. Cache hit rate without avoided work is not useful; latency reduction without answer preservation is not success.

13 Research Lineage and Influence

The field can be read as a sequence of boundary shifts. Memory Networks moved relevant facts outside the recurrent state. Transformer-XL moved context across segments; Block-Recurrent Transformer, FAM, Infini-attention, and Titans made the carried state progressively more structured or adaptive. REALM, RAG, and Atlas moved updateable knowledge outside parameters, while Self-RAG learned when retrieval and critique should occur. RETRO and Memorizing Transformer moved retrieved information closer to hidden-state computation. Unlimiformer and Landmark Attention recast attention as latent token or block retrieval. PagedAttention, Quest, adaptive retention, depth compression, and CacheBlend made allocation, bandwidth, and reuse separate systems problems. RAPTOR, MemWalker, and PageIndex restored document hierarchy. ReAct, IRCoT, Toolformer, and WebGPT made information acquisition an action policy. PRA's proposed step is to make *named reference resolution and reusable latent materialization* a first-class attention capability.

This lineage also tempers novelty claims. Cross-attention over external KV is not new; retrieval during model computation is not new; sparse top-k selection is not new; and caches are not new. PRA's distinct contribution is the package and interface: URI-level identity, explicit reference syntax, separation of naming and materialization, per-layer cache objects, progressive triggering, and a causal evaluation doctrine. Its scientific value depends on showing that this package creates measurable advantages unavailable from simpler combinations.

14 Design Extensions Derived from the Survey

The survey suggests several combinations, but none should be read as a measured PRA capability. Each extension below states a mechanism and the observation that would make it scientifically useful.

14.1 Hierarchical Progressive Retrieval Attention

Represent a document as a URI tree whose nodes expose child handles and optional routing metadata. In the repository example, a project node leads to a module, then a class, and finally a method chunk. A controller first ranks coarse nodes and expands children only when its uncertainty or expected utility warrants the additional work. This proposal combines structural navigation with PRA's detailed token K/V transport. It should be tested against flat chunk routing at matched total gist counts and materialized-token budgets; otherwise a gain could come merely from evaluating more candidates.

14.2 Agent-distilled routing

Use a capable search agent to generate trajectories containing queries, selected sources, section paths, and evidence spans, then convert those trajectories into routing supervision. The deployed model would approximate repeated, read-only search patterns, while the agent remains an escalation path when no advertised object is adequate. The central risk is distilling the agent's shortcuts and index bias. Evaluation must therefore include unseen repositories and compare both task quality and the number of avoided tool calls.

14.3 Multi-fidelity object caches

For each URI, maintain tiers such as metadata, a routing gist, compressed K/V, full token K/V, and raw source. A controller chooses the least expensive tier expected to resolve the current uncertainty. FastGen and PyramidKV suggest non-uniform head/layer budgets; MiniCache suggests depth compression; and Quest suggests loading pages only after the query is known. This connects recurrence and compression with K/V memory management, but it also introduces a new attribution problem: a failure may arise from selecting the wrong object or the wrong fidelity. Factorial ablations with oracle object and oracle fidelity choices can separate those causes.

14.4 Two-stage semantic and physical routing

PRA routing and Quest-style page selection answer questions at different scales. The first chooses a URI and chunk using semantic identity; the second chooses physical K/V pages using a query-dependent score bound. A two-stage runtime can therefore avoid scoring every token while also avoiding transfer of every page inside a selected chunk. Landmark Attention provides a simpler learned-block baseline for the same hierarchy. The decisive experiment reports reference, chunk, and page recall together with bytes loaded. Otherwise a semantic routing gain can be erased by loose physical pages, or a fast page selector can silently omit the only causal evidence.

14.5 Reference compilation

Repeated read-only agent or tool operations can be compiled into stable, versioned references. A SQL result, code-symbol graph, dashboard panel, or document query becomes an object with dependency metadata and an invalidation rule. The possible benefit is avoided resolution and encoding work across repeated queries, not simply a higher cache hit count. Experiments must therefore vary update frequency and sharing while checking that stale versions do not silently improve speed at the expense of correctness.

14.6 Causal provenance graph

Record reference selection, cache version, materialized spans, generated claims, and counterfactual effects, then join these records into a claim–evidence graph. Such a graph would help a developer trace a repair suggestion back to a particular method and source version. The strongest edge is not raw attention weight but a reproducible output change under controlled removal or substitution; attention remains a useful diagnostic, not causal proof.

15 Discussion

15.1 The real design choice is the access contract

Debates framed as "long context versus RAG" hide the more useful question: what contract exists between computation and external information? Dense context offers anonymous token positions. RAG offers retrieved passages. agents offer observations from named actions. PRA proposes persistent objects with identity, version, representation, and cache lifecycle. Such contracts determine not only quality but governance, sharing, invalidation, debugging, and cost.

15.2 When PRA should not win

PRA is unlikely to dominate when the entire source is short, every token is densely relevant, hardware kernels make dense attention cheap, or the source is unknown and requires open-ended discovery. It also adds complexity where a simple one-shot retriever is adequate. Negative results in these regimes would clarify the architecture rather than invalidate it.

15.3 When PRA has a plausible advantage

The strongest regime has large, repeatedly used, naturally addressable objects; sparse relevance; many queries over shared sources; stable or versioned content; and tasks where relevance becomes clearer only after intermediate processing. Examples include technical documentation, analytics catalogs, code repositories, longitudinal records, enterprise knowledge, and embodied-agent maps.

15.4 Training difficulty

A new memory branch can be ignored, overused, or used as a generic adapter. Near-zero gate initialization, frozen-base staged training, auxiliary selection losses, corrupted-reference negatives, and curriculum by distance are sensible. However, explicit supervision may bias the model toward the indexer's notion of relevance. End-to-end loss and causal ablations remain necessary.

15.5 Object identity versus semantic similarity

Semantic retrieval answers "what looks relevant?" Explicit addressing answers "which object is being referred to?" Real systems need both. An explicit handle can restrict search to an object namespace; semantic routing can select parts within it; an agent can discover a missing object. Treating one mechanism as universally sufficient is a category error.

15.6 Impact on agents

PRA does not eliminate the agent loop. It can internalize routine read-only actions, reduce repeated tokenization and tool latency, and provide a memory substrate for agent discoveries. The agent retains planning, source discovery, side effects, and exception handling. The long-term architecture is likely layered rather than exclusive.

15.7 Cognitive analogies are design heuristics

Memory terminology invites cognitive analogies, but the mapping must remain limited. Sliding windows resemble a narrow recency-weighted working set; FAM and recurrent slots resemble maintained working state; landmark tokens resemble retrieval cues; Titans models surprise-weighted consolidation; and PRA handles resemble deliberate use of an external address such as a citation or file path. These analogies can suggest experiments about interference, cue quality, forgetting, and progressive recall. They do not show that the architectures reproduce human memory systems. Neural K/V, online parameter updates, and URI resolvers have different learning rules, embodiment, and social context. The analogy is useful only when it yields a falsifiable computational question.

16 Conclusion

Long-context modeling is not one problem. It is a family of access problems spanning compute graphs, recurrent and test-time state, external memory, retrieval, K/V retention and reuse, indexing, runtime allocation, provenance, and action. The seven categories reviewed here form a continuum from eager anonymous tokens to progressive named actions. PRA occupies a middle point: more structured and reusable than flat long context or one-shot RAG, but potentially cheaper and more differentiable than general search agents. Landmark routing, adaptive RAG, Titans, Quest, and CacheBlend sharpen the required comparisons by solving neighboring parts of the same access problem.

The evidence is narrower than the architectural proposal. The current code establishes mechanical feasibility for chunked, layer-specific routing gists, selected token K/V, variable-length batches, optional summaries, and bounded child-first cache construction. Historical v1 experiments establish approximate ordinary-language compatibility and frozen-path preservation, but they do not validate the corrected router and did not show content-selective reference use. A one-seed corrected-router smoke rerun now verifies the pipeline at two and five splits, while its sub-0.001 shuffled penalties leave the causal question open. The next decisive step is therefore not a larger claim but a controlled rerun: fixed reference-required data, matched token and trainable-capacity budgets, oracle and corrupted-content interventions, multiple seeds, and end-to-end cold and warm costs.

If those experiments show that correct chunks cause reproducible gains at a favorable quality–cost point, PRA would merit study as a model-native interface for repeated referential work. If they do not, the access-contract framework still yields a useful result: it identifies whether the failure lies in naming, selection, materialization, fusion, or runtime reuse, and clarifies where dense context, RAG, structural navigation, and agentic search should meet.

Online Resources

PRA source manuscripts used for this tutorial were supplied by the author. Live links for external works are embedded in the bibliography through DOI or arXiv URLs. The PageIndex discussion refers to its public repository and documentation; claims should be independently reproduced.

References

Akari Asai, Zeqiu Wu, Yizhong Wang, Avirup Sil, and Hannaneh Hajishirzi. Self-RAG: Learning to retrieve, generate, and critique through self-reflection. In *International Conference on Learning Representations*, 2024. URL https://arxiv.org/abs/2310.11511.

Ali Behrouz, Peilin Zhong, and Vahab Mirrokni. Titans: Learning to memorize at test time, 2025. URL https://arxiv.org/abs/2501.00663.

Iz Beltagy, Matthew E. Peters, and Arman Cohan. Longformer: The long-document transformer. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing*, 2020. doi:10.18653/v1/2020.emnlp-main.263. URL https://aclanthology.org/2020.emnlp-main.263/.

Amanda Bertsch, Uri Alon, Graham Neubig, and Matthew R. Gormley. Unlimiformer: Long-range transformers with unlimited length input. In *Advances in Neural Information Processing Systems*, 2023. URL https://arxiv.org/abs/2305.01625.

Sebastian Borgeaud, Arthur Mensch, Jordan Hoffmann, Trevor Cai, Eliza Rutherford, Katie Millican, George B. M. van den Driessche, Jean-Baptiste Lespiau, Bogdan Damoc, Aidan Clark, Diego de Las Casas, Aurelia Guy, Jacob Menick, Roman Ring, Tom Hennigan, Saffron Huang, Loren Maggiore, Chris Jones, Albin Cassirer, Andy

Brock, Michela Paganini, Geoffrey Irving, Oriol Vinyals, Simon Osindero, Karen Simonyan, Jack W. Rae, Erich Elsen, and Laurent Sifre. Improving language models by retrieving from trillions of tokens. In *Proceedings of the 39th International Conference on Machine Learning*, 2022. URL https://proceedings.mlr.press/v162/borgeaud22a.html.

Aydar Bulatov, Yuri Kuratov, and Mikhail S. Burtsev. Recurrent memory transformer, 2022. URL https://arxiv.org/abs/2207.06881.

Zefan Cai, Yichi Zhang, Bofei Gao, Yuliang Liu, Tianyu Liu, Keming Lu, Wayne Xiong, Yue Dong, Baobao Chang, Junjie Hu, and Wen Xiao. PyramidKV: Dynamic KV cache compression based on pyramidal information funneling, 2024. URL https://arxiv.org/abs/2406.02069.

Howard Chen, Ram Pasunuru, Jason Weston, Asli Celikyilmaz, Gheorghe Comanici, and Thibault Févry. MemWalker: Walking through long-context memory with an LLM, 2023. URL https://arxiv.org/abs/2310.05029.

Krzysztof Choromanski, Valerii Likhosherstov, David Dohan, Xingyou Song, Andreea Gane, Tamás Sarlós, Peter Hawkins, Jared Davis, Afroz Mohiuddin, Lukasz Kaiser, David Belanger, Lucy Colwell, and Adrian Weller. Rethinking attention with performers. In *International Conference on Learning Representations*, 2021. URL https://arxiv.org/abs/2009.14794.

Zihang Dai, Zhilin Yang, Yiming Yang, Jaime Carbonell, Quoc V. Le, and Ruslan Salakhutdinov. Transformer-XL: Attentive language models beyond a fixed-length context. In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, 2019. doi:10.18653/v1/P19-1285. URL https://aclanthology.org/P19-1285/.

Darren Edge, Ha Trinh, Newman Cheng, Joshua Bradley, Alex Chao, Apurva Mody, Steven Truitt, Dasha Metropolitansky, Robert Osazuwa Ness, and Jonathan Larson. From local to global: A graph RAG approach to query-focused summarization, 2024. URL https://arxiv.org/abs/2404.16130.

Suyu Ge, Yunan Zhang, Liyuan Liu, Minjia Zhang, Jiawei Han, and Jianfeng Gao. Model tells you what to discard: Adaptive KV cache compression for LLMs, 2023. URL https://arxiv.org/abs/2310.01801.

Alex Graves, Greg Wayne, Malcolm Reynolds, Tim Harley, Ivo Danihelka, Agnieszka Grabska-Barwińska, Sergio Gómez Colmenarejo, Edward Grefenstette, Tiago Ramalho, John Agapiou, Adrià Puigdomènech Badia, Karl Moritz Hermann, Yori Zwols, Georg Ostrovski, Adam Cain, Helen King, Christopher Summerfield, Phil Blunsom, Koray Kavukcuoglu, and Demis Hassabis. Hybrid computing using a neural network with dynamic external memory. *Nature*, 538:471–476, 2016. doi:10.1038/nature20101. URL https://doi.org/10.1038/nature20101.

Kelvin Guu, Kenton Lee, Zora Tung, Panupong Pasupat, and Ming-Wei Chang. REALM: Retrieval-augmented language model pre-training. In *Proceedings of the 37th International Conference on Machine Learning*, 2020. URL https://proceedings.mlr.press/v119/guu20a.html.

DeLesley Hutchins, Imanol Schlag, Yuhuai Wu, Ethan Dyer, and Behnam Neyshabur. Block-recurrent transformers. In *Advances in Neural Information Processing Systems*, 2022. URL https://proceedings.neurips.cc/paper_files/paper/2022/hash/d6e0bbb9fc3f4c10950052ec2359355c-Abstract-Conference.html.

Dongseong Hwang, Weiran Wang, Zhuoyuan Huo, Khe Chai Sim, and Pedro Moreno Mengibar. TransformerFAM: Feedback attention is working memory, 2024. URL https://arxiv.org/abs/2404.09173.

Gautier Izacard, Patrick Lewis, Maria Lomeli, Lucas Hosseini, Fabio Petroni, Timo Schick, Jane Dwivedi-Yu, Armand Joulin, Sebastian Riedel, and Edouard Grave. Atlas: Few-shot learning with retrieval augmented language models, 2022. URL https://arxiv.org/abs/2208.03299.

Urvashi Khandelwal, Omer Levy, Dan Jurafsky, Luke Zettlemoyer, and Mike Lewis. Generalization through memorization: Nearest neighbor language models. In *International Conference on Learning Representations*, 2020. URL https://arxiv.org/abs/1911.00172.

Woosuk Kwon, Zhuohan Li, Siyuan Zhuang, Ying Sheng, Lianmin Zheng, Cody Hao Yu, Joseph E. Gonzalez, Hao Zhang, and Ion Stoica. Efficient memory management for large language model serving with PagedAttention. In *Proceedings of the 29th Symposium on Operating Systems Principles*, 2023. doi:10.1145/3600006.3613165. URL https://doi.org/10.1145/3600006.3613165.

Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Kuttler, Mike Lewis, Wen-tau Yih, Tim Rocktäschel, Sebastian Riedel, and Douwe Kiela. Retrieval-augmented generation for knowledge-intensive NLP tasks. In *Advances in Neural Information Processing Systems*, 2020. URL https://arxiv.org/abs/2005.11401.

Yuhong Li, Yingbing Huang, Bowen Yang, Bharat Venkitesh, Acyr Locatelli, Hanchen Ye, Tianle Cai, Patrick Lewis, and Deming Chen. SnapKV: LLM knows what you are looking for before generation. In *Advances in Neural Information Processing Systems*, 2024. URL https://arxiv.org/abs/2404.14469.

Akide Liu, Jing Liu, Zizheng Pan, Yefei He, Gholamreza Haffari, and Bohan Zhuang. MiniCache: KV cache compression in depth dimension for large language models, 2024. URL https://arxiv.org/abs/2405.14366.

Zichang Liu, Aditya Desai, Fangshuo Liao, Weitao Wang, Victor Xie, Zhaozhuo Xu, Anastasios Kyrillidis, and Anshumali Shrivastava. Scissorhands: Exploiting the persistence of importance hypothesis for LLM KV cache compression at test time. In *Advances in Neural Information Processing Systems*, 2023. URL https://arxiv.org/abs/2305.17118.

Anna Martin-Boyle, Cara A. C. Leckey, Martha C. Brown, and Harmanpreet Kaur. PaperTrail: A claim-evidence interface for grounding provenance in LLM-based scholarly Q&A. In *Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems*, 2026. URL https://arxiv.org/abs/2602.21045.

Amirkeivan Mohtashami and Martin Jaggi. Landmark attention: Random-access infinite context length for transformers, 2023. URL https://arxiv.org/abs/2305.16300.

Tsendsuren Munkhdalai, Manaal Faruqui, and Siddharth Gopal. Leave no context behind: Efficient infinite context transformers with infini-attention, 2024. URL https://arxiv.org/abs/2404.07143.

Reiichiro Nakano, Jacob Hilton, Suchir Balaji, Jeff Wu, Long Ouyang, Christina Kim, Christopher Hesse, Shantanu Jain, Vineet Kosaraju, William Saunders, Xu Jiang, Karl Cobbe, Tyna Eloundou, Gretchen Krueger, Kevin Button, Matthew Knight, Benjamin Chess, and John Schulman. WebGPT: Browser-assisted question-answering with human feedback, 2021. URL https://arxiv.org/abs/2112.09332.

Ofir Press, Muru Zhang, Sewon Min, Ludwig Schmidt, Noah A. Smith, and Mike Lewis. Measuring and narrowing the compositionality gap in language models, 2022. URL https://arxiv.org/abs/2210.03350.

Jack W. Rae, Anna Potapenko, Siddhant M. Jayakumar, and Timothy P. Lillicrap. Compressive transformers for long-range sequence modelling. In *International Conference on Learning Representations*, 2020. URL https://arxiv.org/abs/1911.05507.

Parth Sarthi, Salman Abdullah, Aditi Tuli, Shubh Khanna, Anna Goldie, and Christopher D. Manning. RAPTOR: Recursive abstractive processing for tree-organized retrieval. In *International Conference on Learning Representations*, 2024. URL https://arxiv.org/abs/2401.18059.

Timo Schick, Jane Dwivedi-Yu, Roberto Dessì, Roberta Raileanu, Maria Lomeli, Eric Hambro, Luke Zettlemoyer, Nicola Cancedda, and Thomas Scialom. Toolformer: Language models can teach themselves to use tools. In *Advances in Neural Information Processing Systems*, 2023. URL https://arxiv.org/abs/2302.04761.

Jiaming Tang, Yilong Zhao, Kan Zhu, Guangxuan Xiao, Baris Kasikci, and Song Han. QUEST: Query-aware sparsity for efficient long-context LLM inference. In *Proceedings of the 41st International Conference on Machine Learning*, pages 47901–47911, 2024. URL https://proceedings.mlr.press/v235/tang24l.html.

Harsh Trivedi, Niranjan Balasubramanian, Tushar Khot, and Ashish Sabharwal. Interleaving retrieval with chain-of-thought reasoning for knowledge-intensive multi-step questions. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics*, 2023. doi:10.18653/v1/2023.acl-long.557. URL https://aclanthology.org/2023.acl-long.557/.

Jason Weston, Sumit Chopra, and Antoine Bordes. Memory networks. In *International Conference on Learning Representations*, 2015. URL https://arxiv.org/abs/1410.3916.

Qingyang Wu, Zhenzhong Lan, Kun Qian, Jing Gu, Alborz Geramifard, and Zhou Yu. Memformer: A memory-augmented transformer for sequence modeling, 2020. URL https://arxiv.org/abs/2010.06891.

Yuhuai Wu, Markus N. Rabe, DeLesley Hutchins, and Christian Szegedy. Memorizing transformers. In *International Conference on Learning Representations*, 2022. URL https://arxiv.org/abs/2203.08913.

Guangxuan Xiao, Yuandong Tian, Beidi Chen, Song Han, and Mike Lewis. Efficient streaming language models with attention sinks, 2023. URL https://arxiv.org/abs/2309.17453.

Zichao Yang, Diyi Yang, Chris Dyer, Xiaodong He, Alex Smola, and Eduard Hovy. Hierarchical attention networks for document classification. In *Proceedings of NAACL-HLT*, 2016. doi:10.18653/v1/N16-1174. URL https://aclanthology.org/N16-1174/.

Jiayi Yao, Hanchen Li, Yuhan Liu, Siddhant Ray, Yihua Cheng, Qizheng Zhang, Kuntai Du, Shan Lu, and Junchen Jiang. CacheBlend: Fast large language model serving for RAG with cached knowledge fusion, 2024. URL https://arxiv.org/abs/2405.16444.

Shunyu Yao, Jeffrey Zhao, Dian Yu, Nan Du, Izhak Shafran, Karthik Narasimhan, and Yuan Cao. ReAct: Synergizing reasoning and acting in language models. In *International Conference on Learning Representations*, 2023. URL https://arxiv.org/abs/2210.03629.

Manzil Zaheer, Guru Guruganesh, Avinava Dubey, Joshua Ainslie, Chris Alberti, Santiago Ontañón, Philip Pham, Anirudh Ravula, Qifan Wang, Li Yang, and Amr Ahmed. Big Bird: Transformers for longer sequences. In *Advances in Neural Information Processing Systems*, 2020. URL https://arxiv.org/abs/2007.14062.

Mingtian Zhang, Yu Tang, and PageIndex Team. PageIndex: Next-generation vectorless, reasoning-based RAG. PageIndex blog and open-source repository, 2025. URL https://github.com/VectifyAI/PageIndex/commit/7592163e2a376b3917181fff9ac1858dc5daa2c6. Repository commit 7592163e2a376b3917181fff9ac1858dc5daa2c6; accessed 3 August 2026.

Zhenyu Zhang, Ying Sheng, Tianyi Zhou, Tianlong Chen, Lianmin Zheng, Ruisi Cai, Zhao Song, Yuandong Tian, Christopher Ré, Clark Barrett, Zhangyang Wang, and Beidi Chen. H_2O: Heavy-hitter oracle for efficient generative inference of large language models. In *Advances in Neural Information Processing Systems*, 2023. URL https://arxiv.org/abs/2306.14048.